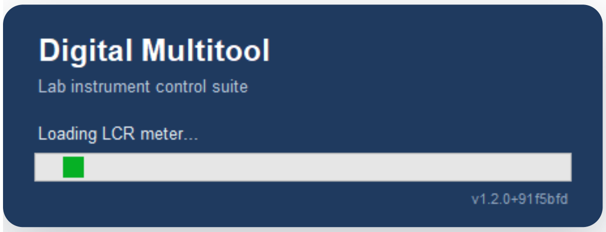
DIGITAL MULTITOOL · V1.2.0

Digital Multitool

User manual for the lab bench-control app — one window for the LCR meter, oscilloscope, signal generator, DC supply, DMM, data logging, battery post-processing, webcam captures and automated SLDEA tests.



CONTENTS

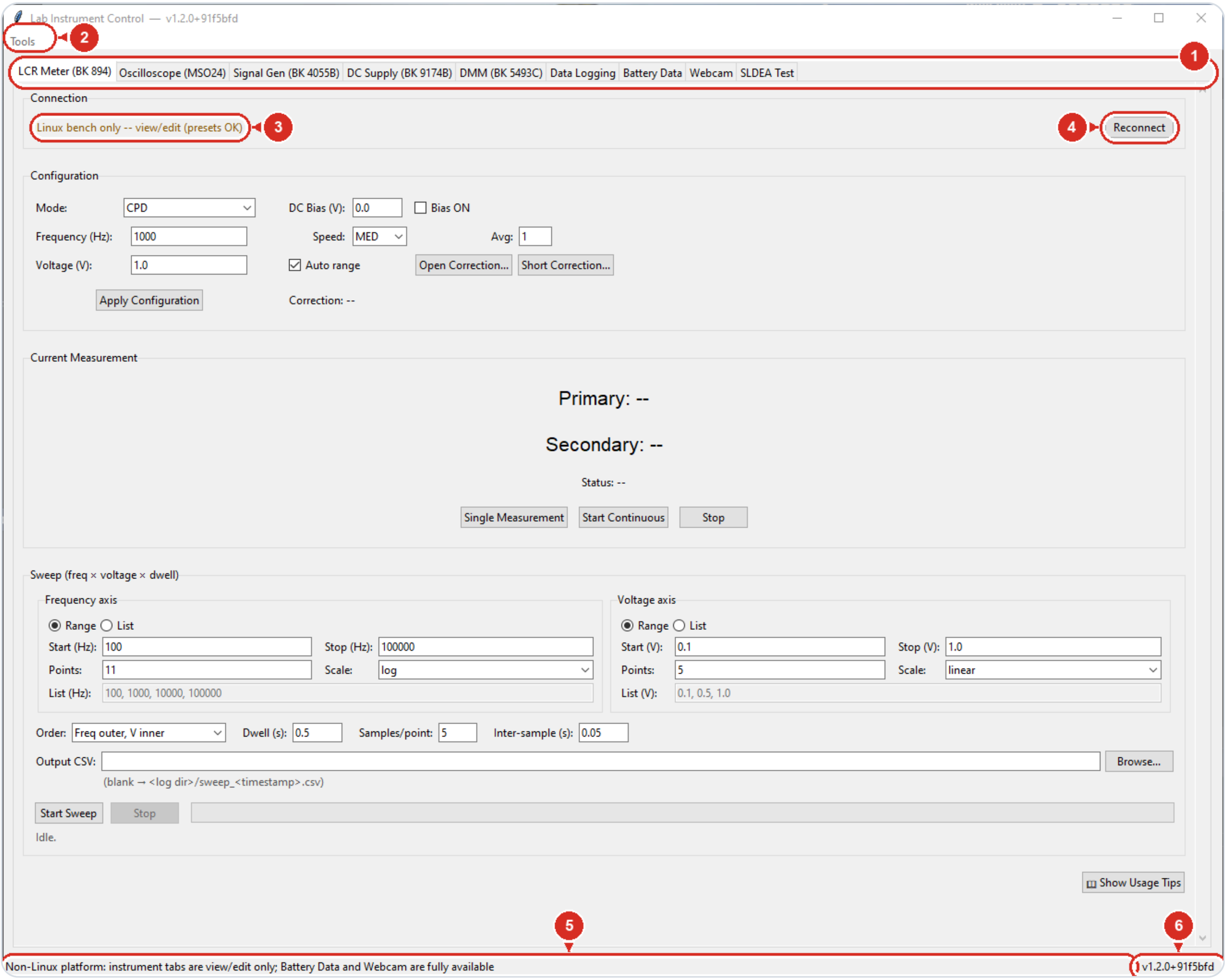
Tip: this PDF has the same chapters in the bookmarks sidebar of your PDF viewer — keep it open for one-click navigation.

Getting started

The frame around the instrument tabs: the launch splash, each tab's connection line with its one-click Reconnect, the Tools menu (whole-bench profiles and the software updater), and the footer status bar with the exact build version.

LAUNCHING

- **Bench PC (Linux):** double-click the desktop icon, or run `.venv/bin/python gui.py`.
- **Windows lab PC:** double-click `_software\Launch_SCPI_Control_Windows.bat` on the ShareDrive. First run needs Python 3.10+ and internet.
- On Windows the instrument tabs are **view/edit only** (amber note in each tab) — Battery Data and Webcam (snapshots and timelapse; not sig-gen-driven sweeps) are fully functional, and presets and bench profiles can be prepared and shared.



1 One tab per instrument or job — click to switch

2 Tools menu — bench profiles & software update

3 Connection status — green once the instrument answers (amber on Windows)	4 Re-detects this one instrument after power-on or replug
5 Status bar — confirmations and warnings appear here	6 Exact version — quote it when reporting a problem

TYPICAL USE

1. Double-click the launcher; the Digital Multitool splash names each tab as it loads.
2. The main window opens and all instruments auto-connect in the background.
3. Check each tab's Connection line turns green with the instrument's identity.
4. If a line is red, power that instrument on and click its Reconnect.
5. Tools > Load Bench Profile... to recall a saved whole-bench setup.
6. Work; watch the footer status bar for confirmations and warnings.
7. Close the window when done — sig-gen outputs and LCR bias switch off automatically; the DC supply output is left as-is.

WATCH OUT

> Closing the window switches off sig-gen outputs and LCR bias and stops a running SLDEA test (best-effort, a few seconds to ramp) — but the DC supply output is never touched. End a live HV run with ■ Abort, not the close button.

> Loading a bench profile immediately rewrites every connected instrument's settings — sig-gen outputs are never switched, but the LCR DC bias follows the profile's Bias ON state. Don't load one mid-measurement.

> On Windows the amber “Linux bench only” connection lines and the greyed-out Update Software... are normal, not faults.

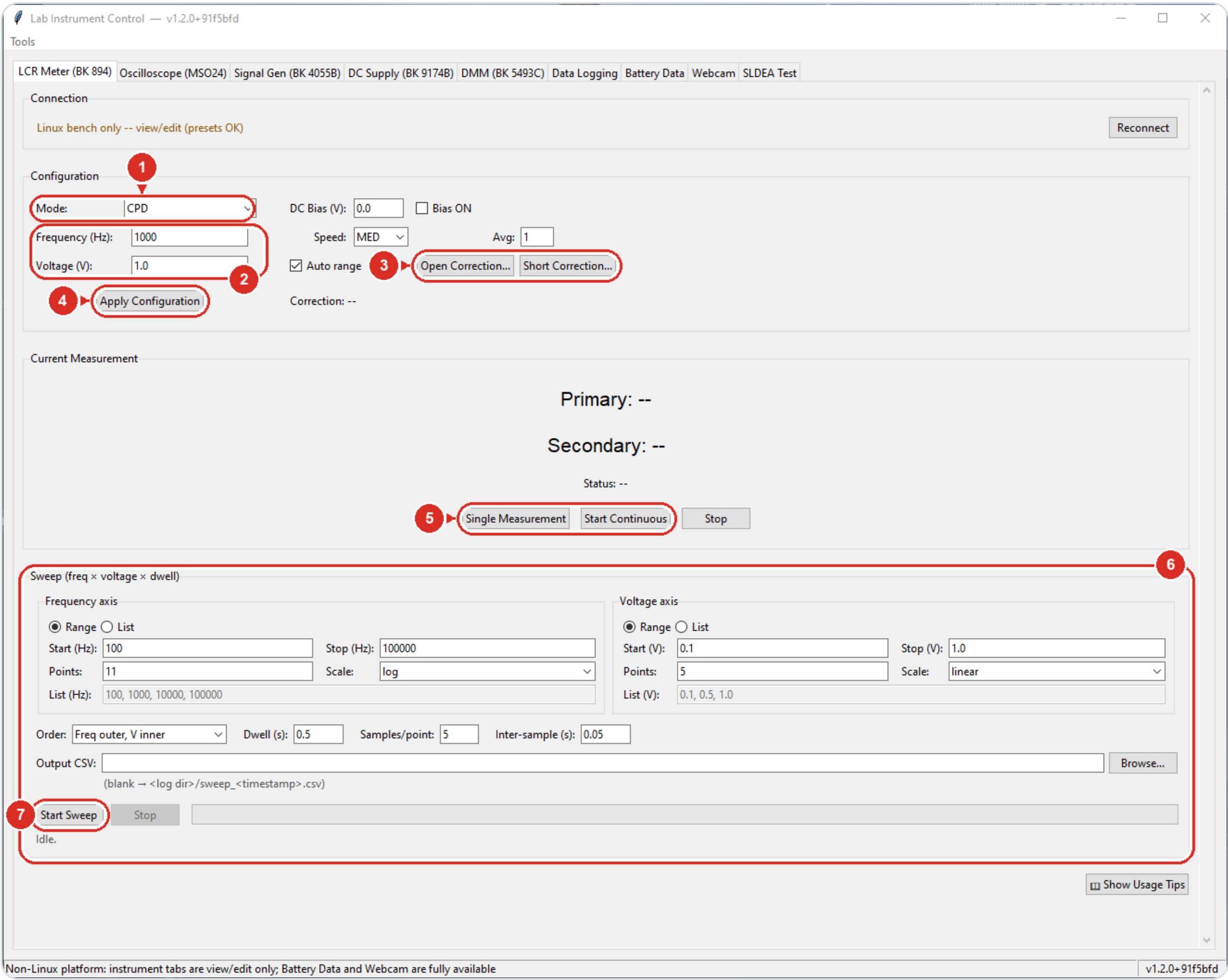
▼ The app shell, menu and profiles in detail

Splash screen (Digital Multitool)	Appears instantly at launch and names each tab as it builds (a few seconds), then closes itself.
Window title	Shows the app name plus the exact version and build hash you are running.
Connection line (top of each instrument tab)	Green 'Connected (USB/LAN/Serial)' with the instrument's identity, red 'Not connected' with the reason, amber 'Linux bench only' on Windows.
Reconnect (top-right of each instrument tab)	Closes and reopens that one instrument; use after powering a box on or replugging its cable.

Status bar (footer, left)	Running commentary: save/load confirmations, busy notices, and shared-drive fallback warnings.
Version readout (footer, right)	Version plus build hash; quote it when reporting a problem.
Tools > Save Bench Profile...	Names and stores a snapshot of the whole bench (LCR, scope, both sig-gen channels) in the shared library.
Tools > Load Bench Profile...	Restores a named snapshot into the fields and pushes it to whatever is connected; outputs are never switched.
Tools > Delete Bench Profile...	Removes a named profile after a confirmation.
Tools > Save Bench Profile to File... / Load Bench Profile from File...	Exports or imports a single profile as a .json file for a USB stick or email.
Tools > Update Software...	Pulls the latest release, redeploys the shared-drive copy for all users, then offers Restart now (Linux bench only).
Shared preset storage	Presets and profiles live in the ShareDrive presets folder so both bench users see them; if the NAS is down, saves land in a local copy and the status bar says so.

LCR Meter — BK 894

Controls the B&K Precision 894 LCR meter: set the AC test signal (mode, frequency, level, DC bias), take single or live readings of a component, and run automated frequency-by-voltage sweeps that save to a CSV file.



1. Measurement pair — CPD = capacitance + dissipation
2. AC test signal — meter clamps at 2.0 V and 500 kHz
3. Fixture zeroing — redo both after changing fixture or leads
4. Nothing reaches the meter until you click this
5. One reading now, or live readings every 0.2 s
6. Automated frequency × voltage sweep — define both axes here
7. Runs the sweep in the background → CSV file

TYPICAL USE

1. Check Connection is green; click Reconnect if not.
2. Pick Mode, then set Frequency and Voltage.
3. Optional: run Open then Short Correction on the empty/shorted fixture.
4. Click Apply Configuration.
5. Connect the part; click Single Measurement or Start Continuous.
6. Read Primary/Secondary; Status OK means a good reading.
7. For curves: set the sweep axes and click Start Sweep — results go to CSV.

WATCH OUT

> The app enforces the meter's limits — voltage 0.01–2.0 V, frequency 100 Hz–500 kHz; out-of-range entries are rejected with a Configuration Error on Apply.

> Nothing reaches the meter until Apply Configuration; the Primary/Secondary readouts are labeled by the last APPLIED mode, not whatever the dropdown currently shows.

> DC bias stays on until you untick Bias ON and click Apply — do not leave bias across a bias-sensitive part.

> Open/Short corrections belong to the current fixture and leads — redo both after changing either, or every reading carries the wrong offset.

> Each correction sweep takes tens of seconds and the meter is busy until it finishes.

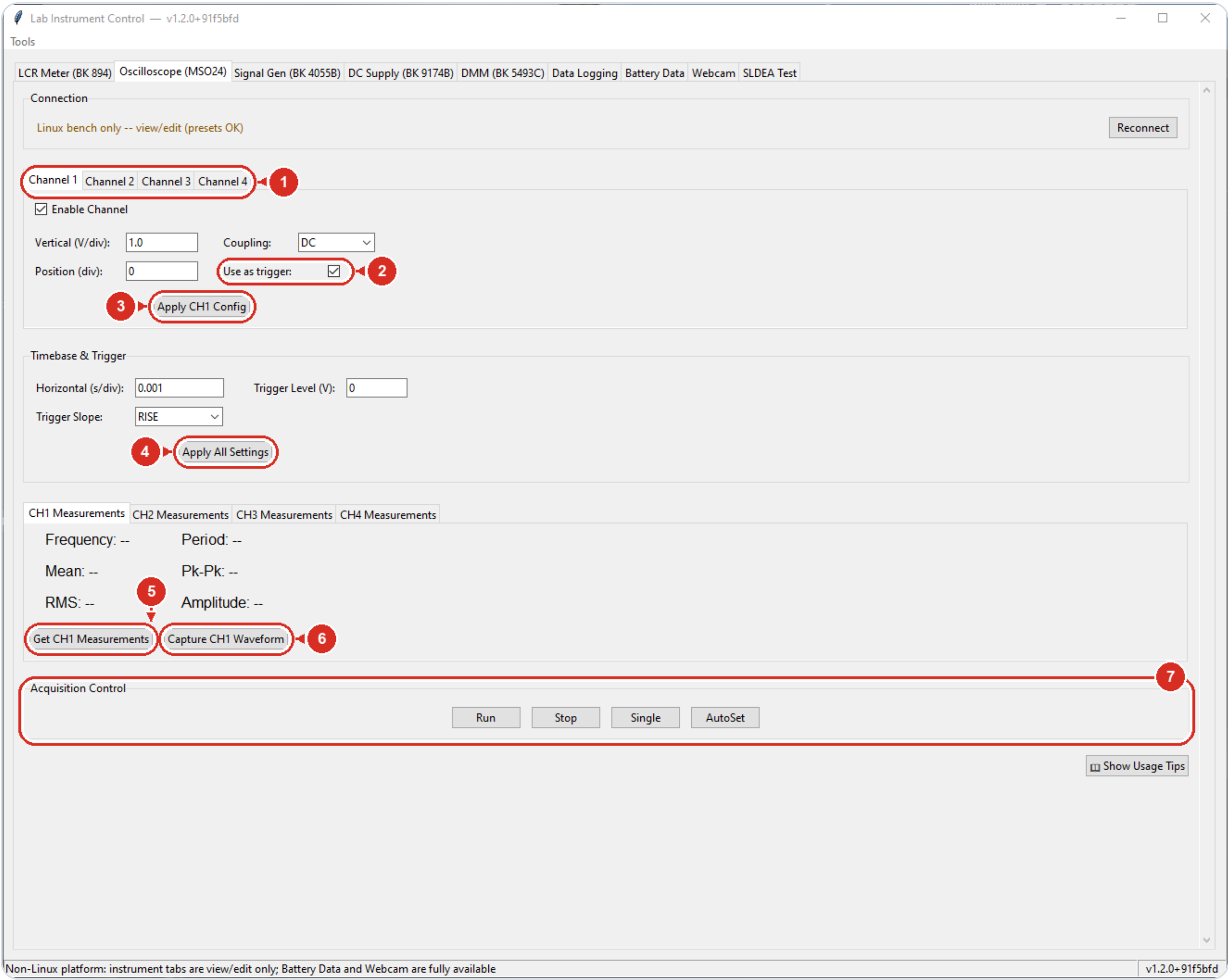
▼ Every control on this tab

Connection / Reconnect	Shows meter link state (red = not connected); Reconnect re-detects the meter and reads its current settings into the tab.
Mode	Picks the measurement pair (e.g. CPD = capacitance + dissipation); CP/LP are parallel-model, CS/LS series-model variants.
Frequency (Hz)	AC test frequency, 100 Hz to 500 kHz; 1 kHz is the standard capacitor test, 100 Hz for electrolytics.
Voltage (V)	AC test level, 0.01 to 2.0 V; 1.0 V is standard and the meter clamps anything higher to 2.0 V.
DC Bias (V) + Bias ON	Steady DC voltage held across the part during the AC test (for C-vs-bias curves); active only while Bias ON is ticked, pushed on Apply.
Speed / Avg	Reading integration time (SLOW = quietest, FAST = quickest) and how many raw readings (1-256) the meter averages into each result.

Auto range	ON lets the meter pick its range; turn OFF to hold the current range and avoid mid-sweep range-change glitches on a fixed part.
Open Correction… / Short Correction…	Fixture de-embedding sweeps (fixture empty / terminals shorted); the label beside them shows which corrections are currently ON.
Apply Configuration	Pushes everything above to the meter — no setting takes effect until this is clicked.
Current Measurement (Primary / Secondary / Status)	Latest reading, labeled by the mode actually applied to the meter; Status OK means a valid measurement.
Single Measurement / Start Continuous / Stop	One reading now, live readings every 0.2 s, or halt the live readout.
Frequency axis / Voltage axis (Range | List)	Defines each sweep axis either as start-stop-points (linear or log) or as an explicit comma-separated list.
Order / Dwell (s) / Samples/point / Inter-sample (s)	Which axis is the outer loop, settling wait after each setpoint change, readings taken per point, and pause between them.
Output CSV / Browse…	Where the sweep file goes; left blank it lands in the log directory as sweep_<timestamp>.csv.
Start Sweep / Stop / progress bar	Runs the sweep in the background with live progress; Stop finishes the current sample and keeps everything written so far.

Oscilloscope — Tektronix MSO24

Controls the Tektronix MSO24 4-channel oscilloscope: set up each channel, timebase and trigger, start/stop acquisition, read automated measurements, and pull a waveform record into a plot window for saving as CSV.



1. Per-channel setup — one inner tab per channel (CH1–CH4)
2. Tick the channel that should trigger acquisition
3. Sends this channel only — does not touch the trigger
4. The only button that sends trigger source, level and slope
5. Readouts are snapshots — click to refresh
6. Pulls the trace into a plot window with Save CSV
7. Run / Stop / Single / AutoSet

TYPICAL USE

1. Check Connection shows connected; click Reconnect if it is red.
2. On each Channel tab: tick Enable Channel, set V/div, Coupling, Position.
3. Tick Use as trigger on the channel you want to trigger from.
4. Set Horizontal (s/div), Trigger Level (V) and Trigger Slope.
5. Click Apply All Settings (or AutoSet for an unknown signal).
6. Click Run, or Single for one triggered event.
7. On the channel's Measurements tab, click Get CHx Measurements; use Capture CHx Waveform to plot and Save CSV.

WATCH OUT

> Typed values do nothing until you click an Apply button — and only Apply All Settings sends the trigger source/level/slope; Apply CHx Config does not touch the trigger.

> AutoSet changes scale, timebase and trigger on the instrument but the fields on this tab do NOT update to match — clicking Apply All Settings afterwards will overwrite what AutoSet chose.

> Measurement readouts are snapshots — they only update when you click Get CHx Measurements. Frequency/Period show "No signal" on non-periodic input.

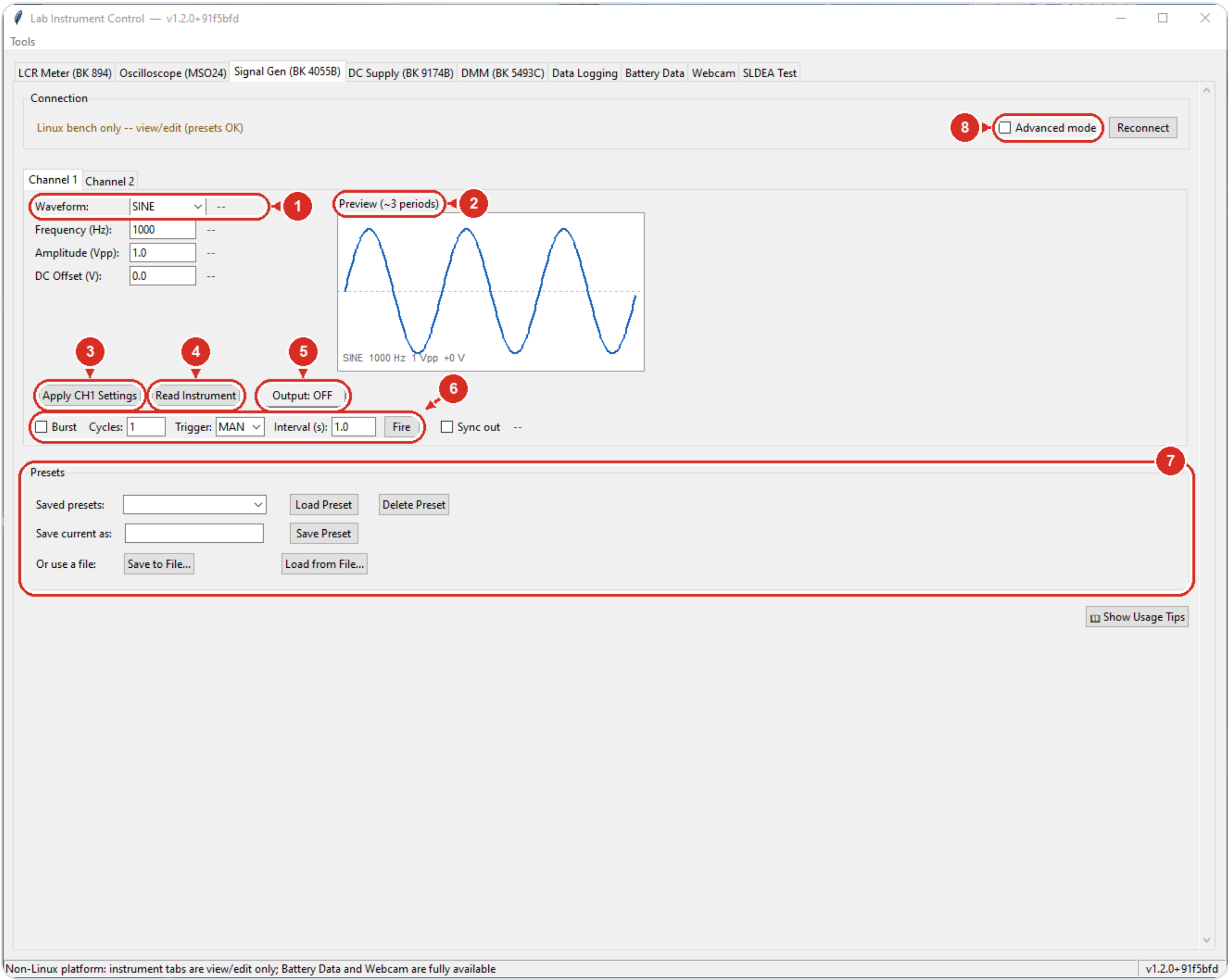
▼ Every control on this tab

Connection / Reconnect	Shows scope link status (red = not connected); Reconnect retries auto-detection over USB.
Channel 1–4 tabs	One settings page per input channel; the tab you are on is the channel you are editing.
Enable Channel	Turns the channel display on or off on the scope immediately.
Vertical (V/div)	Voltage per grid division for this channel; aim for the signal filling ~60–80% of the screen.
Coupling	DC shows everything, AC blocks the DC level (ripple on a bias), GND gives a flat zero reference.
Position (div)	Moves the trace up or down, in divisions from center.
Use as trigger	Marks this channel as the edge-trigger source; if several are ticked, the lowest number wins.
Apply CH1 Config	Sends this one channel's enable, scale, position and coupling to the scope (one such button per channel tab).
Horizontal (s/div)	Timebase — seconds per division; whole screen is 10 divisions. Show 2–3 cycles of a periodic signal.
Trigger Level (V)	Voltage the trigger-source signal must cross to start an acquisition; set it mid-signal for a stable display.
Trigger Slope	Trigger on the RISE (rising) or FALL (falling) edge.

Apply All Settings	Pushes every channel plus timebase, trigger source, level and slope in one go — the only button that sets the trigger source.
CH1–CH4 Measurements tabs	Frequency, Period, Mean, Pk-Pk, RMS and Amplitude readouts per channel; Get CHx Measurements refreshes them, Capture CHx Waveform opens the sample record in a plot window with Save CSV.
Run / Stop / Single	Run acquires continuously, Stop freezes the display, Single arms one triggered capture then stops.
AutoSet	Lets the scope pick scale and trigger for an unknown signal — good starting point, then fine-tune.

Signal Generator — BK 4055B

Controls both output channels of the BK 4055B function generator: waveform type, frequency, amplitude, offset, burst shots, and the physical output switch. Named presets let you save and recall complete two-channel setups.



1. Waveform type — the fields below adapt to it
2. Drawing of your typed settings — not a measurement
3. Pushes settings; never switches the output
4. Pull the front-panel state back into the fields
5. The real output switch — asks first, green while ON
6. Burst mode — exactly N cycles per trigger; Fire sends one
7. Named presets store both channels; files travel by USB
8. Reveals phase, pulse edges, load and polarity

TYPICAL USE

1. Pick the Channel 1 or Channel 2 inner tab.
2. Choose a Waveform; the fields adapt to it.
3. Enter frequency, amplitude and offset; check the preview.
4. Click Apply CH1 Settings and compare the gray applied readouts.
5. Click Output: OFF and confirm to energize the output.
6. For shot testing: tick Burst, set Cycles, Apply, then Fire.
7. Click the green Output button to switch off when done.

WATCH OUT

> Turning Output ON can energize the Trek HV amplifier: 1 V from this generator = 1 kV at the DEA. The confirmation defaults to No — check amplitude and offset first.

> If a gray applied readout differs from what you typed, the instrument clamped or coerced your value — believe the gray number.

> Amplitude is calibrated for the selected Load (Advanced mode). With Load set to 50 Ω but the output actually open/High-Z, the real voltage is about twice what you asked for.

> Custom (ARB) waveform data cannot be sent over USB — the instrument's firmware caps USB commands at 52 bytes and larger transfers freeze it until a power cycle. The app blocks such uploads; use LAN or the Waveform Editor's flash-drive .bin route instead.

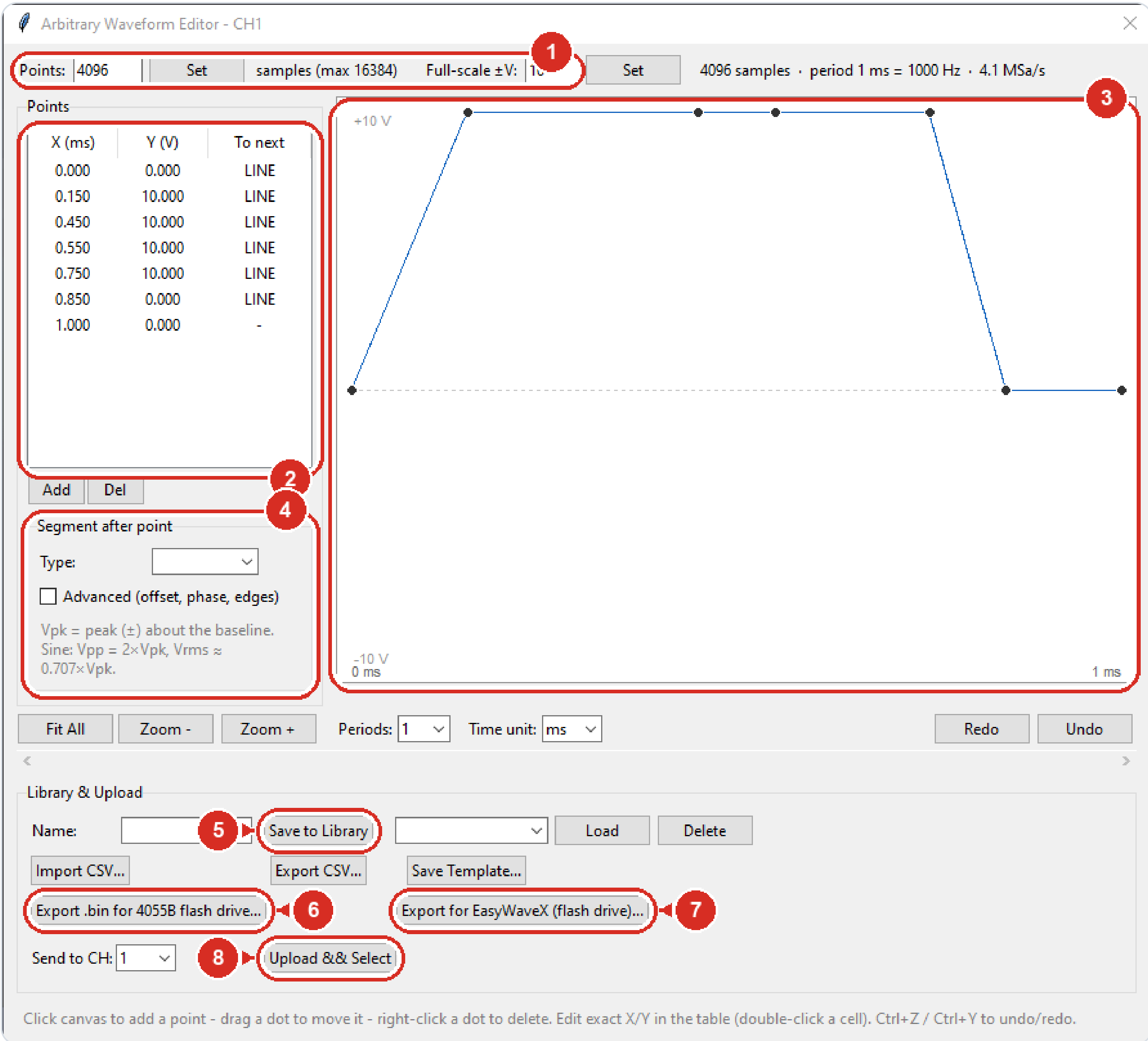
▼ Every control on this tab

Reconnect	Retry the connection to the generator (tries LAN first, then USB).
Advanced mode	Reveals the extra fields: Phase, pulse Rise/Fall/Delay, Load and Polarity; Apply pushes them even when hidden.
Channel 1 / Channel 2	Inner tabs — each output is configured independently.
Waveform	Pick SINE, SQUARE, RAMP, PULSE, NOISE, DC or ARB; the fields below change to match.
Frequency / Amplitude / DC Offset (and per-waveform fields)	Type values; the gray text beside each field shows what the instrument actually accepted after Apply.
Waveform Editor...	Shown only when Waveform = ARB — opens the arbitrary-waveform designer window; the label beside it names the chosen arb.
Preview (~3 periods)	Live drawing of your typed-in settings — it is not a measurement.

Apply CH1 Settings / Apply CH2 Settings	Push the configuration to the instrument; never switches the output on or off.
Read Instrument	Pull the generator's current state back into the fields — use after changing things on the front panel.
Output: OFF / ON	Switches the real output; asks for confirmation before turning ON and goes green while ON.
Burst + Cycles / Trigger / Interval	Emit exactly N cycles per trigger, then idle — bounded energy per shot; MAN = Fire button, INT = auto-repeat, EXT = rear Aux In.
Fire	Sends one manual burst trigger (needs Burst on, trigger MAN, output ON).
Sync out	Puts a trigger edge on the rear Sync BNC every period — feed it to the scope for stable triggering.
Save Preset / Load Preset / Delete Preset	Named presets store BOTH channels; Load fills the fields and pushes the configuration to the instrument (output state untouched).
Save to File… / Load from File…	Export/import a preset as a .json file so it can travel on a USB stick or live in a project folder.

Arbitrary Waveform Editor

Design a custom waveform for the BK 4055B signal generator by placing points on a plot and choosing the shape between them. Save it to the app's library, then deliver it to the instrument by flash-drive file (.bin or EasyWaveX CSV) or by LAN upload. Opened from the Signal Gen tab: set a channel's waveform to ARB and click "Waveform Editor...".



1. Resolution (samples) and the volt scale (max ±10 V)
2. Breakpoint table — double-click a cell for exact values
3. Click to add a point, drag dots, right-click deletes
4. Shape to the next point: LINE, HOLD, SINE, EXP...

5 Store the waveform for reuse	6 Preferred delivery: file for the 4055B's front USB port
7 Fallback: CSV template for the EasyWaveX PC	8 Direct upload — works over LAN only

TYPICAL USE

1. On the Signal Gen tab, set the channel waveform to ARB and click "Waveform Editor...".
2. Set Points, Full-scale ±V, and the Time unit.
3. Click the plot to add points and drag the dots into shape.
4. Select a point and pick the segment type (and parameters) to the next point.
5. Check the header readout — the X span is one period, which sets the output frequency.
6. Type a Name and click Save to Library.
7. Click "Export .bin for 4055B flash drive..." and save to a USB stick (or "Upload && Select" if on LAN).

WATCH OUT

> The export buttons only save a file — nothing is sent to the instrument. The .bin holds the shape only; the EasyWaveX CSV embeds frequency/amp/offset in its header.

> After recalling the .bin on the 4055B front panel, click Apply CHx Settings on the Signal Gen tab to push frequency, amplitude and offset (the export pre-fills them).

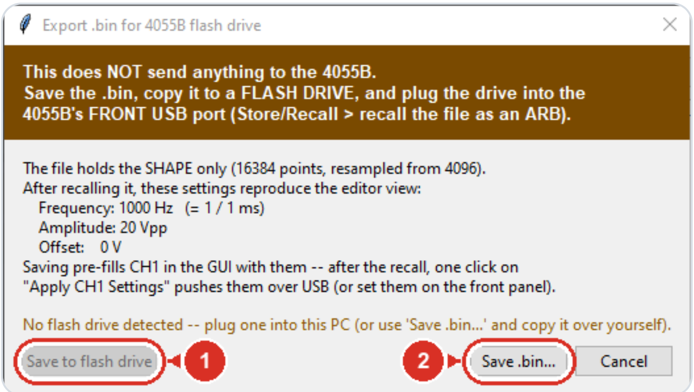
> "Upload && Select" works only over a LAN connection. Over USB the upload is refused (the 4055B's 52-byte firmware cap can wedge the box) — use the flash-drive .bin route instead. With no instrument connected (e.g. the Windows viewing copy) it reports "not connected"; designing and exporting still work.

> Closing the editor with unsaved edits discards them after one confirmation — anything not saved to the library is gone.

▼ Every control on this tab

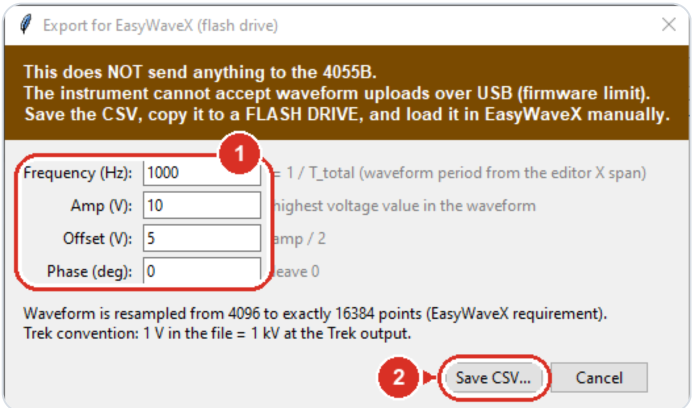
Points + Set	Total samples in the waveform (resolution), 2 to 16384.
Full-scale ±V: + Set	Volts at the top/bottom of the plot; capped at ±10 V (4055B limit).
Header readout	Live summary: samples, period (= X span), derived frequency, sample rate.
Points table	Lists every breakpoint; double-click an X or Y cell to type an exact value.
Add / Del	Insert a point mid-interval after the selected row, or delete the selected point.
Segment after point	Shape from the selected point to the next: LINE, HOLD, SINE, SQUARE, RAMP, PULSE, or EXP, with per-shape parameters.

Advanced (offset, phase, edges)	Reveals the extra segment parameters (offset, phase, rise/fall).
Preview plot	Click empty space to add a point, drag a dot to move it, right-click a dot to delete it.
Fit All / Zoom - / Zoom + / scrollbar	Zoom and pan the preview; Fit All shows the whole waveform.
Periods / Time unit	Preview 1-3 repeats; time unit (µs/ms/s) sets what X means — the X span is one period, so it sets the output frequency.
Undo / Redo	Step edits back/forward (Ctrl+Z / Ctrl+Y, 50 levels).
Name + Save to Library	Store the waveform (re-editable) in the app library under this name.
Library list + Load / Delete	Recall a saved waveform into the editor, or remove it from the library.
Import CSV... / Export CSV... / Save Template...	Bring in or write out the raw sample list; Template writes an example CSV to fill in.
Send to CH + Upload && Select	Push the waveform to the chosen channel over LAN and select it (needs a LAN-connected 4055B).



1 One click straight onto a detected USB stick

2 ...or choose any folder yourself

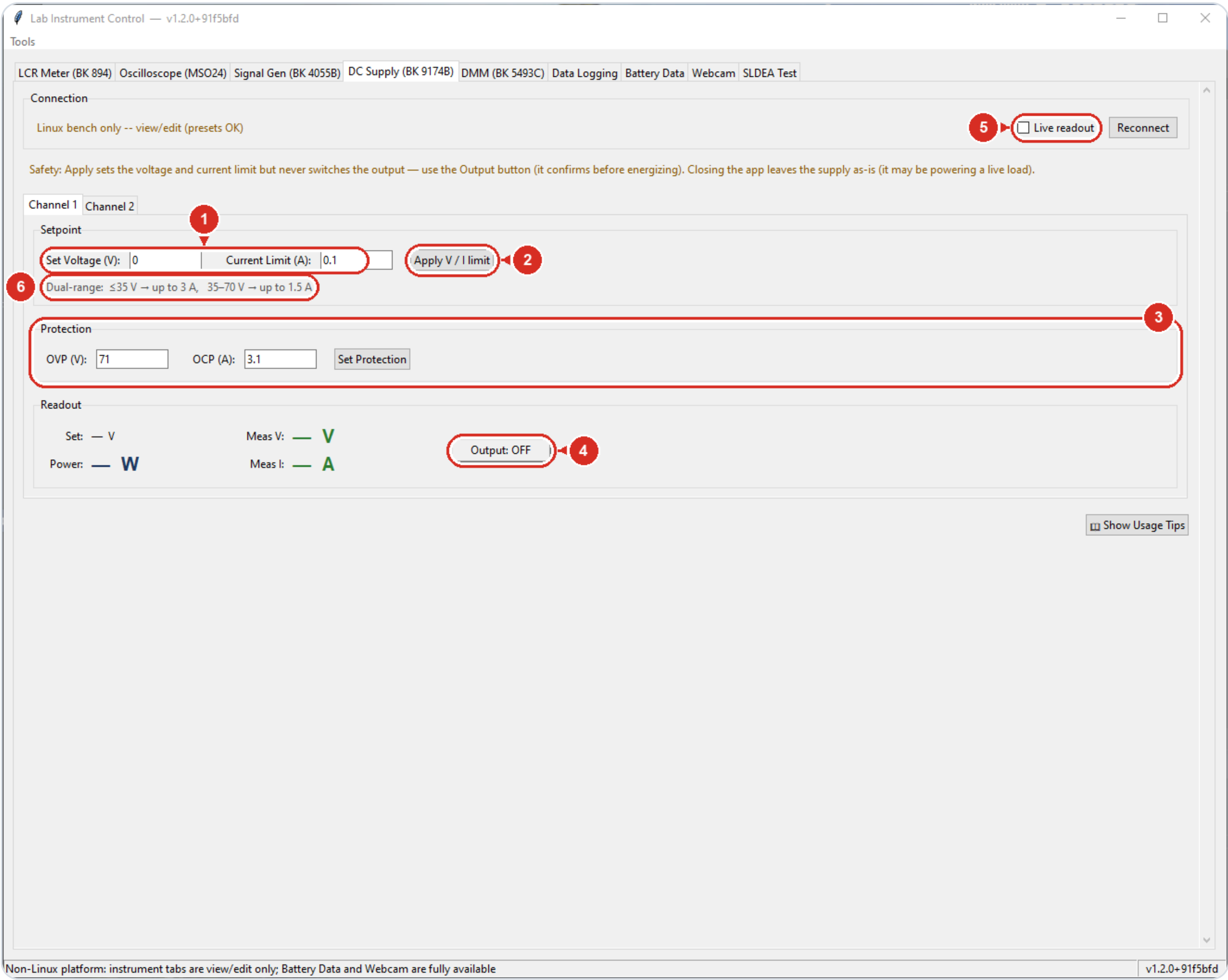


1 Header values — lab defaults pre-filled (1 V = 1 kV at the Trek)

2 Writes the exact template EasyWaveX expects

DC Supply — BK 9174B

Controls the B&K 9174B dual-output bench DC supply: per channel you stage a voltage and current limit, set hardware protection trips, watch live measured volts/amps/watts, and switch the output on or off. Setting values and energizing the terminals are deliberately separate actions.



1. Stage voltage and current ceiling (CC — not a fuse)
2. Sends values — output stays untouched
3. OVP / OCP hardware trips — set these first
4. Energizes the terminals — confirms first, green while ON
5. Polls measured V, A and W every 0.5 s
6. Allowed envelope: 3 A up to 35 V, 1.5 A above

TYPICAL USE

1. Check Connection is not red; press Reconnect if needed.
2. Pick the Channel 1 or Channel 2 sub-tab.
3. Enter OVP and OCP, press Set Protection.
4. Enter Set Voltage and Current Limit, press Apply V / I limit.
5. Press the Output button and confirm — it turns green ON.
6. Tick Live readout to watch measured V, A, and W.
7. Press the green Output button again to switch off when done.

WATCH OUT

> Apply V / I limit never switches the output — only the Output button energizes the terminals, and it asks first.

> Closing the app leaves the supply exactly as-is: an ON output keeps powering the load.

> The current limit is constant-current, not a fuse — at the limit the supply lowers voltage to hold the current, so set it just above the expected draw.

> Above 35 V each output is capped at 1.5 A; Apply refuses voltage/current pairs outside the dual-range envelope.

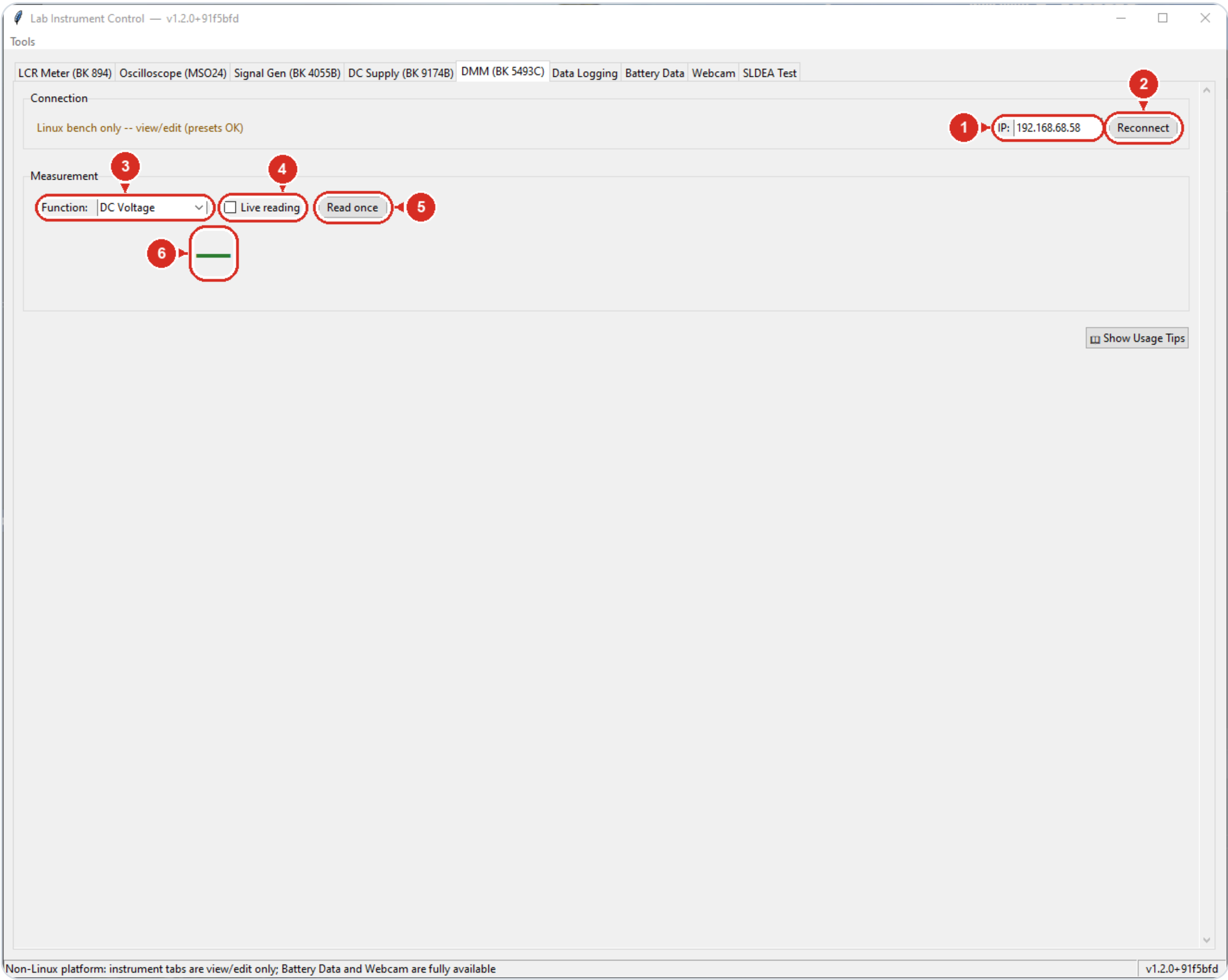
▼ Every control on this tab

Connection / Reconnect	Shows link status (red = not connected); Reconnect reopens the supply's USB-serial port.
Live readout (checkbox)	Polls measured voltage, current, and power on both channels every 0.5 s (read-only).
Channel 1 / Channel 2 (sub-tabs)	Pick which of the two independent outputs the panel below controls.
Set Voltage (V)	Voltage to program on this channel (0–70 V); takes effect on Apply.
Current Limit (A)	Constant-current ceiling for this channel; the supply drops voltage to hold it.
Apply V / I limit	Sends voltage and current limit to the channel — never touches the output switch.
Dual-range note	Allowed combinations: up to 3 A at ≤35 V, only 1.5 A from 35–70 V; Apply refuses values outside this.
OVP (V) / OCP (A)	Over-voltage / over-current hardware trip points (defaults 71 V / 3.1 A).
Set Protection	Writes the OVP/OCP trip points to the channel; do this before enabling output.
Readout (Set / Meas V / Meas I / Power)	Programmed voltage plus measured volts, amps, and computed watts; updates with Live readout on.

Output: OFF / ON (button)	Toggles the channel output; asks for confirmation before energizing and turns green when ON.
📖 **Show Usage Tips**	Pops up a quick-reference summary of this tab.

Digital Multimeter — BK 5493C

Reads the bench B&K 5493C 6.5-digit multimeter over the lab network (this unit is Ethernet-only). Pick a measurement function and take a single reading or watch a live, continuously updated display.



1. The meter's LAN address — this unit is Ethernet-only
2. Connect / retry at this IP
3. What to measure — V, A, Ω, Hz, F (auto-ranged)
4. Continuous display, ~2 readings per second
5. Single measurement
6. Reading appears here — red OVLD on overload

TYPICAL USE

1. On the meter's front panel: Menu -> I/O / LAN, use "DHCP Once" to get an address on the bench subnet.
2. Type that IP into the IP: field and click Reconnect; status turns green.
3. Pick a Function from the dropdown (starts on DC Voltage).
4. Click Read once, or tick Live reading for a continuous ~2 Hz display.
5. Read the big number; the grey line beneath shows the raw value and unit.
6. To record: tick "DMM" on the Data Logging tab — it logs the function selected here to CSV.

WATCH OUT

> This unit is LAN-only — its USB never enumerated (known hardware issue). Do not troubleshoot USB; fix the network instead.

> The meter's socket accepts one client at a time. If Reconnect keeps failing, close any other program (or stale session) talking to it, or power-cycle the meter.

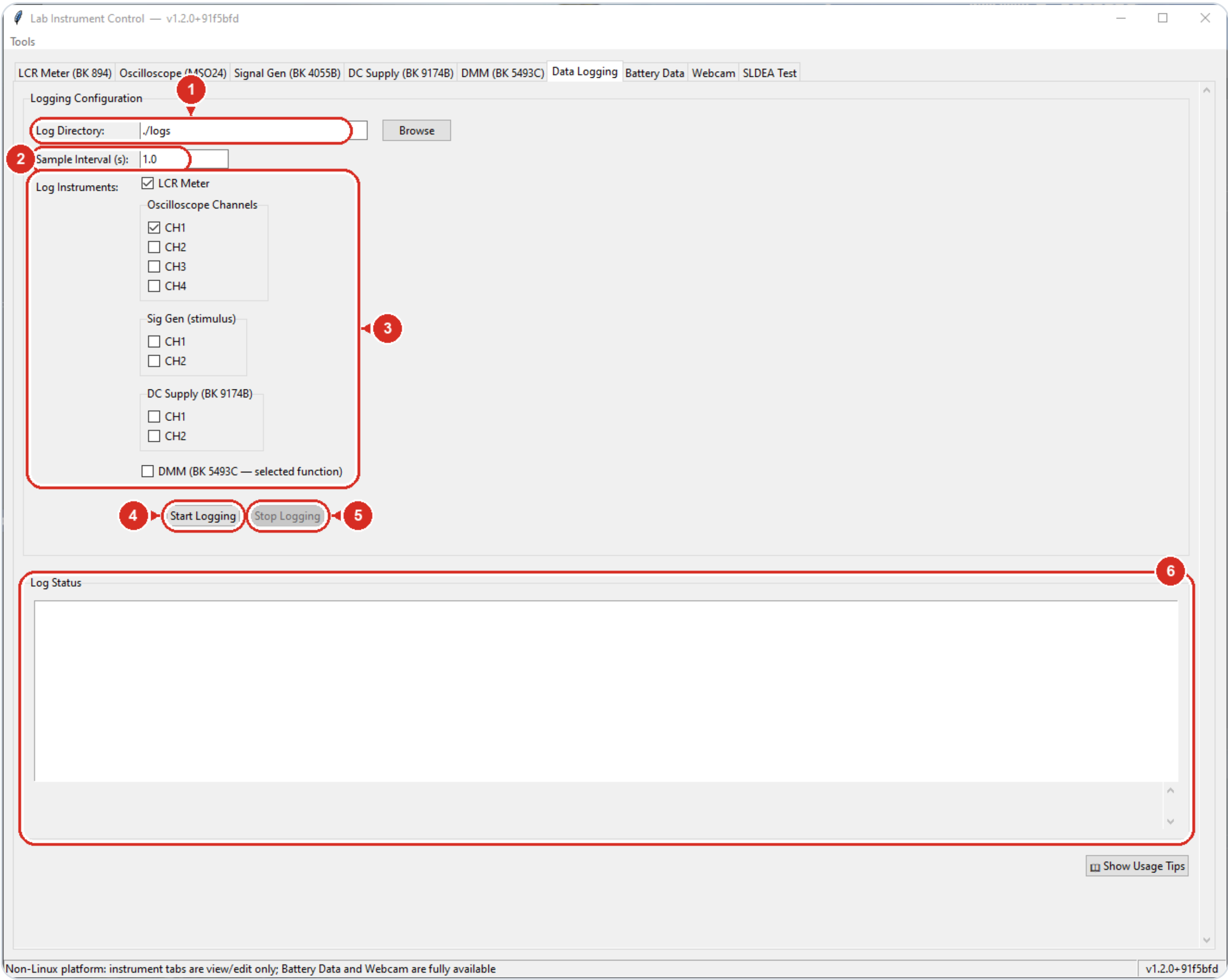
> Data Logging records the Function selected at the moment you start logging; changing the dropdown mid-log does not change what is recorded.

▼ Every control on this tab

Not connected (status)	Shows link state to the meter; red until a connection succeeds.
IP:	The meter's LAN address (default 192.168.68.58); set it on the meter's front panel first.
Reconnect	Connects (or retries) to the meter at the IP shown, on port 45454.
Function:	What to measure: DC/AC Voltage, DC/AC Current, 2W/4W Resistance, Frequency, Capacitance — all auto-ranged.
Live reading	Polls the selected function every 0.5 s (~2 Hz) while ticked.
Read once	Takes a single measurement of the selected function.
Big readout	Latest reading in engineering notation (e.g. 16.74 mV); turns red "OVLD" on overload or a non-numeric reply.
Small line under readout	The function name plus the raw value and unit.
📖 Show Usage Tips	Opens a short reference on connection, use, and logging.

Data Logging

Records readings from the connected instruments to CSV files at a fixed interval — one timestamped file per source, so a run captures stimulus (sig gen, supply) and response (LCR, scope, DMM) side by side. Files are for offline analysis (Excel, Python); there is no live plot on this tab.



1. All CSV files land here (default ./logs)
2. Seconds between samples
3. Tick every source to record — one CSV per source
4. Opens fresh timestamped CSVs and starts sampling
5. Closes the files and ends the run
6. File paths, errors, and dropped sources appear here

TYPICAL USE

1. Connect and configure the instruments on their own tabs first.
2. Pick a Log Directory (Browse) or keep ./logs.
3. Enter the Sample Interval in seconds (e.g. 1.0).
4. Tick every source you want recorded.
5. Click Start Logging and note the CSV paths in Log Status.
6. Watch Log Status during the run for errors or dropped sources.
7. Click Stop Logging, then open the CSVs in Excel or Python.

WATCH OUT

> A source that errors 5 times in a row is silently dropped for the rest of the run (one notice in Log Status) — check Log Status after long runs; if every source dies, logging stops itself.

> The DMM is logged with whatever function was selected on the DMM tab at the moment you pressed Start Logging — changing the function mid-run does not change what is recorded.

> Sources are read one after another each tick, so with many sources or slow instruments the real spacing can be longer than the Sample Interval you typed.

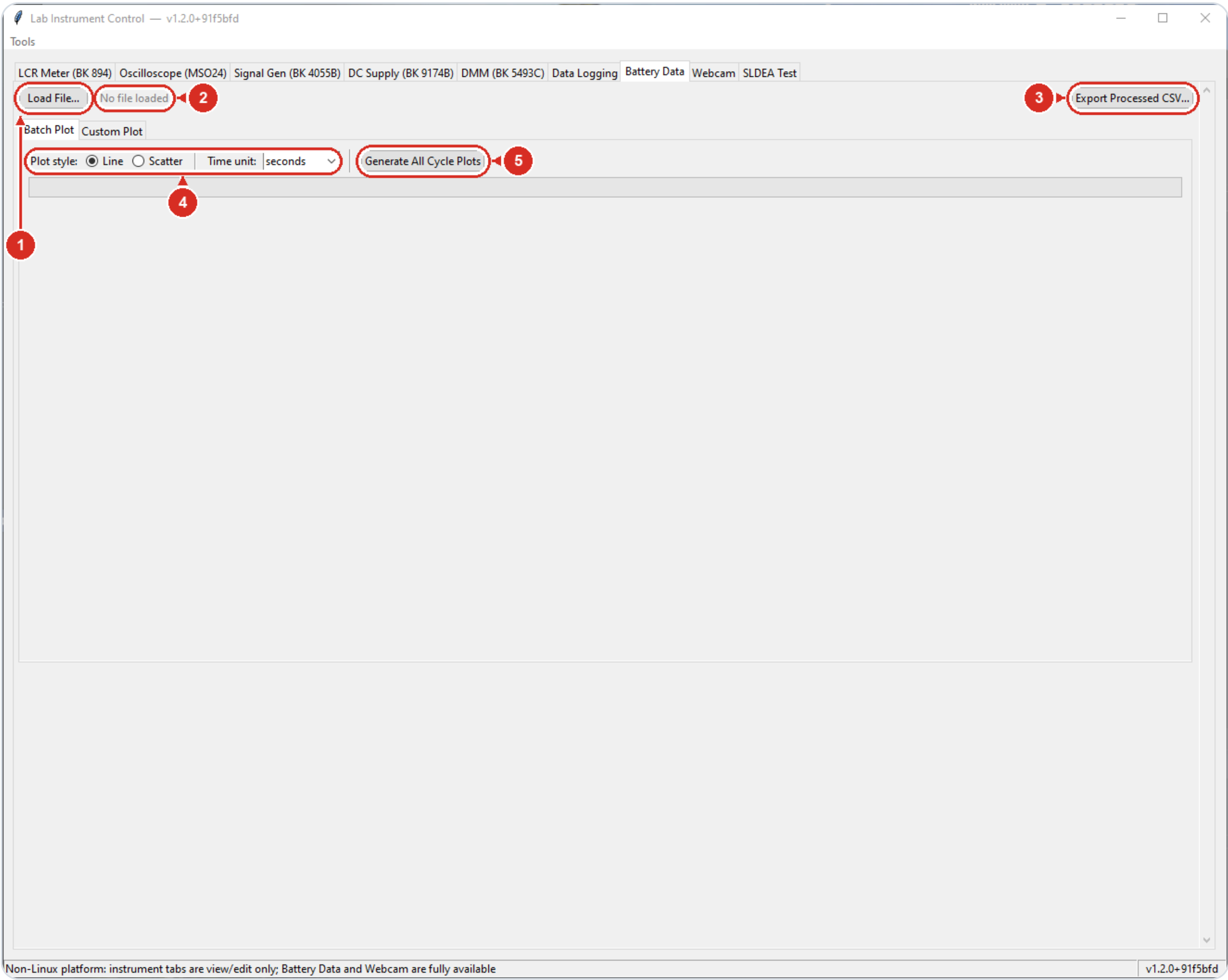
▼ Every control on this tab

Log Directory: + Browse	Folder where the CSV files are written (default ./logs); created automatically if missing.
Sample Interval (s):	Seconds between samples; every ticked source is read once per interval.
LCR Meter	Log the LCR reading each tick: mode, frequency, primary, secondary, status.
Oscilloscope Channels (CH1–CH4)	Log scope measurements per ticked channel: frequency, period, mean, pk-pk, RMS, amplitude.
Sig Gen (stimulus) (CH1/CH2)	Log what was driving the DUT: waveform, frequency, amplitude, offset, output ON/OFF.
DC Supply (BK 9174B) (CH1/CH2)	Log set voltage, measured voltage, measured current, and calculated power per channel.
DMM (BK 5493C — selected function)	Log the function currently selected on the DMM tab (value + unit).
Start Logging	Opens one fresh timestamped CSV per ticked source and begins sampling; sources not connected are skipped with a notice.
Stop Logging	Closes the files and ends the run (takes effect within about a second).
Log Status	Run messages: where each CSV was opened, per-source errors, dropped sources.

📖 **Show Usage Tips**	Popup listing the exact CSV columns per instrument and analysis hints.

Battery Data

Post-processes exports from the battery cycler: load the tester's .xls/.xlsx file and the tab translates the Mandarin headers and status labels to English, merges all detail sheets into one table, and converts mV to V. From there you can export a clean CSV or make plots. Works on any PC — no instrument connection involved.



1. Open the raw cycler .xls/.xlsx export
2. Shows filename + rows × columns when processed
3. Translated, merged table → CSV
4. Batch options — line/scatter, time in s/min/h
5. Two 300-dpi PNGs per cycle (V–t and V–Q)

TYPICAL USE

1. Click Load File... and pick the untouched cycler export (.xls/.xlsx).
2. Wait until the label shows the filename with row and column counts.
3. Optional: click Export Processed CSV... to save the translated table.
4. For per-cycle PNGs: on Batch Plot, set style and time unit, click Generate All Cycle Plots, pick a folder.
5. For one specific plot: on Custom Plot, choose X and Y columns, style, and cycles.
6. Click Preview Plot to check it in the panel.
7. Click Save as PNG (300 dpi)... to export it.

WATCH OUT

> Batch plotting overwrites existing PNGs with the same cycle-numbered names in the chosen folder without asking.

> Batch plots deliberately leave out rest periods; only charge/discharge segments appear. Use Custom Plot with Color by status to see rests.

> Load only the raw tester export: it must keep its original sheet layout (Info first, Cycle summary last). Re-saved or trimmed files fail to load or pull in the wrong sheets.

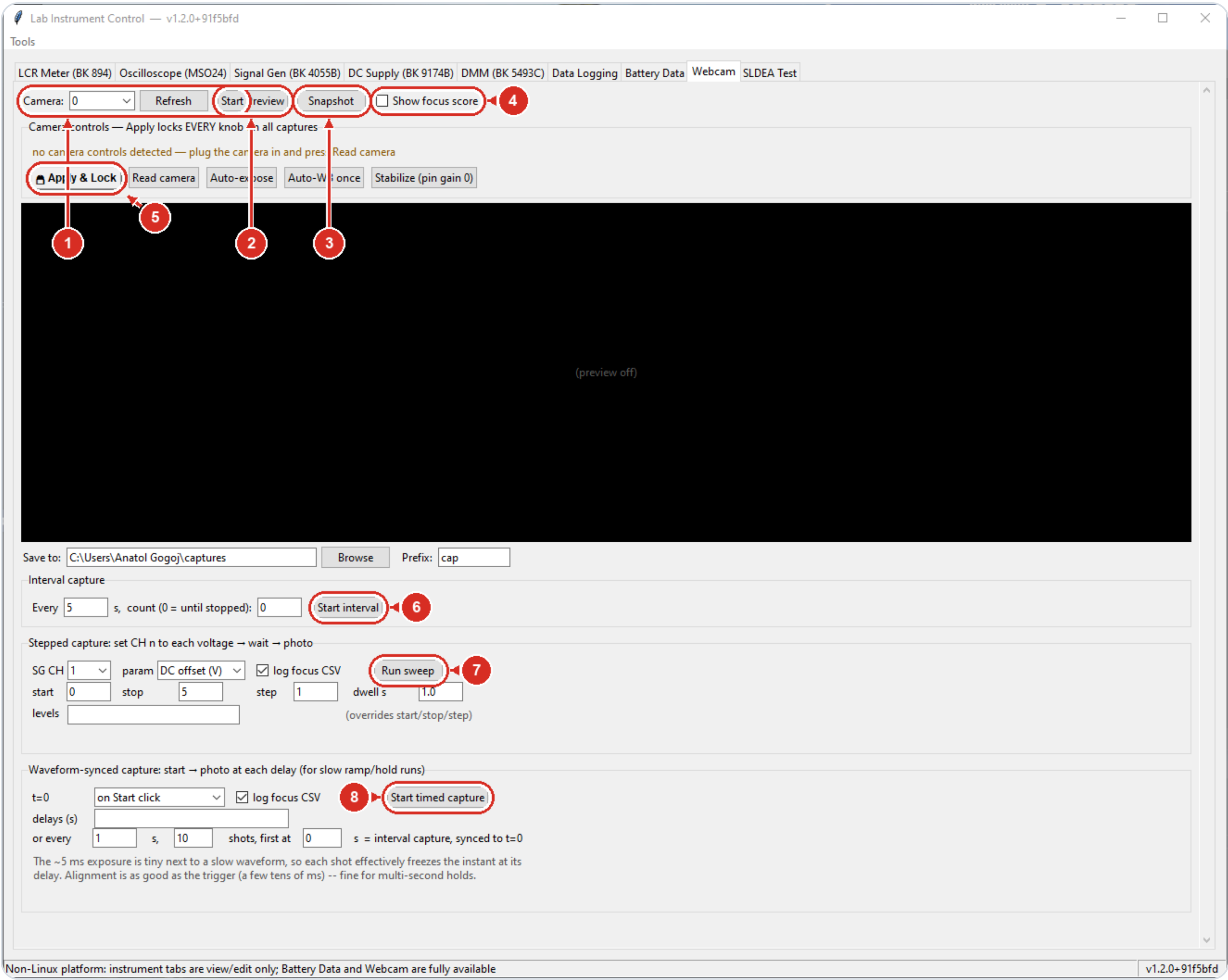
▼ Every control on this tab

Load File...	Opens a cycler .xls/.xlsx export and processes it in the background; the gray label beside it shows the filename plus row and column counts when done.
Export Processed CSV...	Saves the full translated table as a CSV (default name: <file>_processed.csv).
Batch Plot / Custom Plot	Inner tabs that switch between one-click per-cycle plotting and a build-your-own plot panel.
Plot style: Line / Scatter (Batch Plot)	Draw batch plots as connected lines or as dots.
Time unit (Batch Plot)	Seconds, minutes, or hours for the time axis of batch plots.
Generate All Cycle Plots	Asks for an output folder, then saves two 300-dpi PNGs per cycle: voltage vs time and voltage vs capacity.
Progress bar (Batch Plot)	Fills as batch plots are written; the status text ends with a count and the folder.
X Axis / Y Axis (Custom Plot)	Pick any numeric column for each axis; defaults to relative_time_s vs voltage_V.
Style: Line / Scatter (Custom Plot)	Line or dot rendering for the custom plot.
Color by status	Splits the trace into colored segments per charge/discharge/rest state, with a legend.
Time unit (Custom Plot)	Rescales time-based axes to seconds, minutes, or hours.

Cycles: All cycles / Selected:	Plot everything, or type cycle numbers like 1,2,3 or 1-5 in the entry box.
Preview Plot	Draws the plot in the panel on the right; use the toolbar under it to zoom and pan.
Save as PNG (300 dpi)...	Redraws the custom plot and saves it as a 300-dpi PNG.

Webcam

Live view and photo capture from the bench USB/industrial camera. Take single snapshots, timelapse series, or automated runs where each photo is tied to a signal-generator voltage or to a delay after a start trigger — with an on-screen focus score to help you sharpen the lens first.



1. Pick the camera; Refresh rescans devices
2. Live view — the same button stops it
3. Save one timestamped PNG
4. Sharpness number — turn the lens until it peaks
5. Locks every camera knob — prevents drift during runs
6. Automatic photo every N seconds
7. Steps a sig-gen voltage, one photo per level
8. Photos at chosen delays after t=0

TYPICAL USE

1. Pick the camera number and click Start Preview.
2. Tick Show focus score; turn the lens until the number peaks.
3. Set exposure/colour (or use Stabilize), then 🔒 Apply & Lock.
4. Set the save folder and filename prefix.
5. Snapshot for one photo, or Start interval for a timelapse.
6. For voltage-vs-image runs: fill start/stop/step and dwell, click Run sweep.
7. Each filename carries shot number, timestamp, and the level.

WATCH OUT

> Before a sweep, turn the channel's Output ON in the Signal Gen tab — otherwise the levels are written but never reach the device, and you get a folder of identical photos.

> A sweep leaves the generator sitting at the LAST level when it ends — it does not return the output to zero.

> Without 🔒 Apply & Lock the camera's firmware auto-gain/auto-WB drifts brightness and colour mid-run.

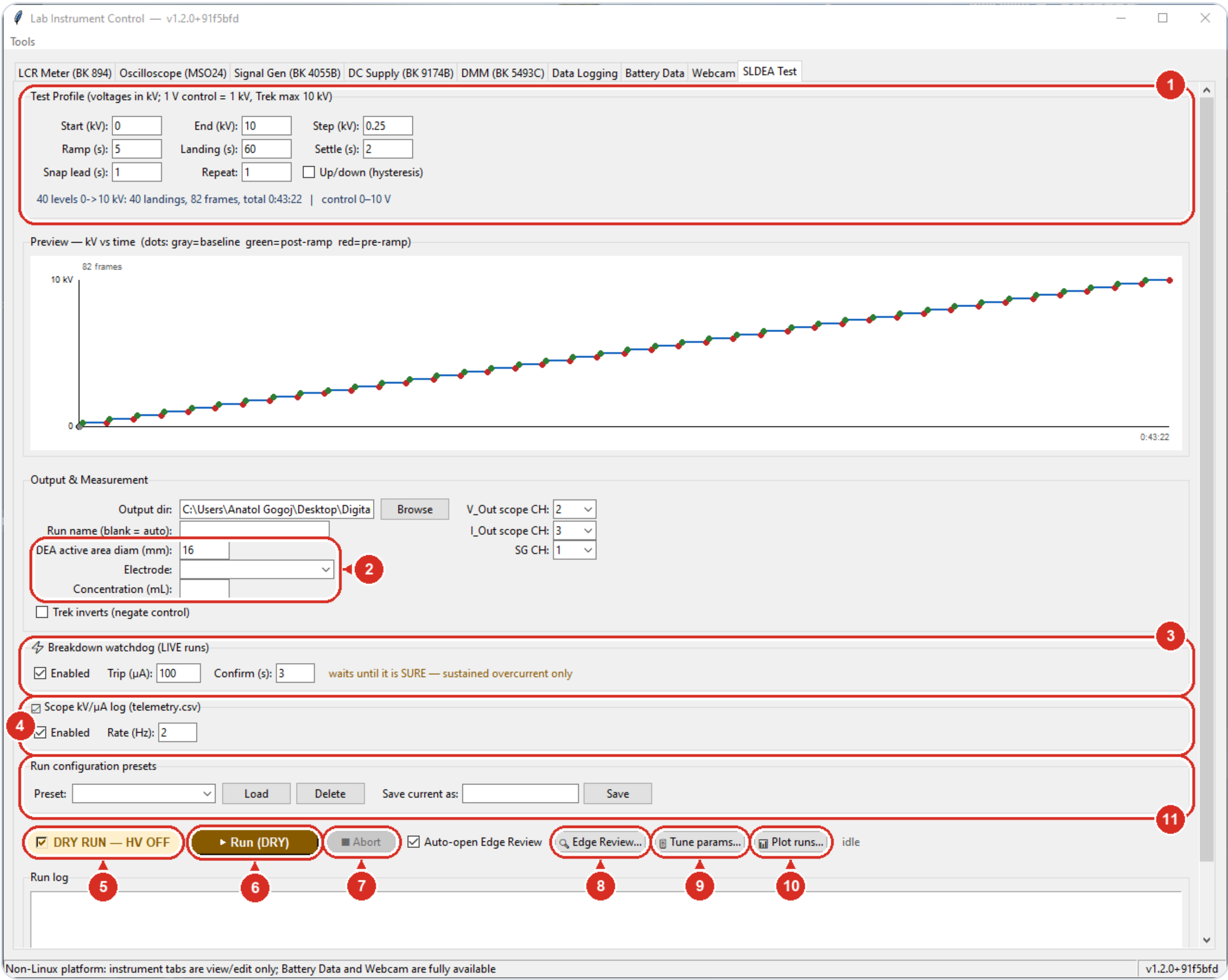
▼ Every control on this tab

Camera + Refresh	Pick the video device number; Refresh rescans after plugging or unplugging a camera.
Start Preview	Live view of the selected camera; the same button stops it.
Snapshot	Saves the current preview frame as a timestamped PNG in the save folder.
Show focus score	Overlays a live sharpness number and a green area-of-interest circle — higher = sharper; turn the lens to maximise it.
Camera controls panel	One field per camera knob (exposure, gain, white balance…), built from what the camera reports.
🔒 Apply & Lock	Writes every panel value to the camera, keeps re-asserting them during all captures, and saves them for the next start.
Read camera	Refills the fields from the camera's current state.
Auto-expose	Searches for an exposure giving a mid-grey image and fills it in; then Apply & Lock.
Auto-WB once	One-shot grey-world colour balance on the current scene; stop the preview first.
Stabilize (pin gain 0)	Pins gain at 0 (defeats the firmware auto-gain) and finds a matching exposure — the only drift-free combo.
Save to / Prefix	Output folder and the start of every saved filename (e.g. cap_0003_20260728-141230_1p9V.png).
Interval capture	Automatic snapshot every N seconds; count 0 keeps going until you press Stop.

Stepped capture (Run sweep)	Sets the chosen sig-gen channel to each voltage (start/stop/step or an explicit list), waits the dwell, saves a photo per level.
log focus CSV	Also writes a CSV with one row per shot: level or delay, filename, sharpness score.
Waveform-synced capture	Photos at chosen delays after t=0 (Start click or a channel burst trigger) — for imaging fixed points of a slow ramp/hold run.

SLDEA Test

Runs an automated voltage-staircase test on a single-layer dielectric elastomer actuator (SLDEA): the signal generator drives the Trek HV amplifier (1 V control = 1 kV, max 10 kV), the scope logs the Trek's V_Out/I_Out monitors, and the webcam photographs the electrode at each voltage landing. Each run produces a self-contained folder (setup.txt, data.csv, run.log, frames/ — plus telemetry.csv when the scope is connected) ready for the Edge Review analysis program.



1 The voltage staircase — start/end/step, ramp and landing times (the 0 kV reference photo is always taken)

2 Defines the device: active-area diameter (sets the px→mm scale — a wrong value corrupts every area), electrode material, ink concentration (greyed for non-inks)

3 Aborts on sustained overcurrent — leave Enabled

4 Logs kV/µA continuously to telemetry.csv — the current between photos

5 Safety toggle — untick only for a live HV run

6 Starts the run — the label shows the mode

7 Ramps to 0 kV first, then stops

8 Open Edge Review on a finished run — see the Edge Review section

9 Advanced — confirmation-gated; Save rewrites the run's detection settings

10 Several finished runs on one figure — Export writes the PNG and its tidy CSV together

11 Save the whole tab under a name and recall it — never the run name, and never the LIVE state

TYPICAL USE

1. Enter the staircase: Start/End/Step kV, Ramp and Landing times.
2. Check the preview and summary line — dots are photos, total time shown.
3. Set Output dir, DEA active area diam (mm), and the scope/SG channels.
4. Do a dry run first to rehearse timing and camera captures.
5. Untick DRY RUN; answer the scope-monitor and Energize HV prompts.
6. In camera pre-flight, center the DEA in the circle, check focus, start.
7. Let it finish (or ■ Abort); Edge Review opens on the new run folder.

WATCH OUT

A live run drives real HV up to 10 kV through the Trek. Stop with ■ Abort (it ramps to 0 kV first); closing the window also stops the run best-effort — if the SG link is dead the Trek can stay energized and the app raises an HV NOT ZEROED alarm.

If the scope-monitor check warns and you pick 'run anyway', the run completes but logs blank measured kV and wrong currents — this silently ruined five runs on 2026-07-25.

A wrong DEA diameter corrupts the px-to-mm scale: every area number from that run is wrong and setup.txt must be hand-fixed before re-analysis.

🎚 Tune params... is an advanced tool: it asks for confirmation first, and its Save rewrites the run's setup.txt — the detection settings every later analysis pass will use.

▼ Every control on this tab

Start / End / Step (kV)	Defines the voltage staircase; End must be 10 kV or below.
Ramp (s) / Landing (s)	Transition time between levels and hold time at each level.
Settle (s) / Snap lead (s)	When the two photos per landing are taken: settle = wait after the ramp before the post-ramp photo; snap lead = how long before the next ramp the pre-ramp photo fires.
Repeat / Up/down (hysteresis)	Repeat the whole sweep N times; Up/down adds a descending leg for hysteresis curves.

0 kV baseline photo	Captured automatically at the start of every run — Edge Review measures all areas against it. (No longer a checkbox: unticking it silently broke every outline.)
Preview — kV vs time	Live plot of the staircase; dots mark photo moments (gray baseline, green post-ramp, red pre-ramp) and a red playhead tracks a running test.
Output dir / Run name (blank = auto)	Where the run folder is created; blank name auto-stamps SLDEA_YYYYMMDD_HHMMSS.
V_Out / I_Out scope CH, SG CH	Which scope channels carry the Trek monitor outputs and which SG channel drives the Trek.
DEA active area diam (mm)	Nominal resting diameter of the active area — written to setup.txt and used as the px-to-mm scale in Edge Review.
Trek inverts (negate control)	Tick when the Trek is configured inverting; the run then drives negative control so the HV output stays positive.
Electrode	Which compliant electrode material this device uses. Pick one of the listed inks — Carbon Solutions P2/P3-SWNT, nano-c Invisicon 3500/3900, plain CNT, carbon black, eGaIn — or type any material straight into the box; whatever you type is recorded word for word in setup.txt, alongside a canonical family so runs can be grouped later. Optional, but the run asks before starting without it. Leave it blank if the sample is genuinely unknown, rather than naming something you are not sure of.
Concentration (mL)	How much CNT ink went on this device — the '2.5mL' in a folder name like P3_2.5mL_Triazole, now recorded in the run itself. It is greyed out for carbon black and eGaIn, which are not inks dispensed by volume: when it is greyed the run neither asks for it nor writes it to setup.txt, so a carbon-black run carries no ink key at all. Where it does apply, the run asks before starting without it, and only a positive number is accepted.
⚡ Breakdown watchdog (LIVE runs)	Watches Trek current during live runs; current above Trip (µA) sustained for Confirm (s) snapshots the breakdown, ramps to 0 and aborts. Transient spikes never trip it.
📈 Scope kV/µA log (telemetry.csv)	Writes the scope’s kV/µA continuously to telemetry.csv beside data.csv, so the current BETWEEN photos is recorded and a breakdown can be dated electrically. Rate (Hz) is capped at 2 — at that setting the current rows are free, because they are the breakdown watchdog’s own readings, which used to be thrown away. Voltage is read less often (at most once a second) because it costs a second scope query; the commanded voltage is on every row regardless. Needs the scope; works on dry runs too. Nothing reads the file — data.csv is still the run, and Edge Review, the tuner and the diagnostic are unaffected.

Run configuration presets	Saves everything above under a name and recalls it later, so a campaign's staircase is not retyped each session. A preset covers the staircase, Output dir, scope/SG channels, DEA diameter, electrode, Trek inverts, and the watchdog and telemetry settings. It deliberately does NOT carry your run name, and never the DRY/LIVE state — loading one always leaves the tab in DRY, so arming the HV stays a separate deliberate click. Presets live in the same shared presets/ library as the signal-generator presets. If a preset was written by an older or newer version and something in it no longer fits, the load says what it could not apply instead of skipping it quietly.
DRY RUN — HV OFF / ▶ Run / ■ Abort	Dry run rehearses timing and photos with HV off; unticking turns the row red for a live run. Abort ramps to 0 kV before stopping.
Auto-open Edge Review	When a run finishes with frames, launches Edge Review on it and starts a default detection pass.
🔍 Edge Review… / 🎚 Tune params…	Open the analysis program, or the live slider tuner. Tune params is advanced and confirmation-gated — its Save rewrites the run's setup.txt.
📊 Plot runs…	Opens the plot window on several finished runs at once: tick the runs, choose area / current / power, toggle what is drawn (pre/post separately, mean line, uncertainty bands, breakdown marks) and the figure redraws as you go. Nothing is written until Export, which saves the 300 dpi PNG together with the tidy per-snapshot CSV behind it — the CSV is the figure's evidence, so the two always travel as a pair. Area mode needs runs Edge Review has saved areas for (✓ processed in the list); current and power work on raw runs, so a run can be looked at electrically before anyone has reviewed a frame. Hover the uncertainty-bands box to see where those numbers come from: they are the calibrated error budget in SLDEA_MEASUREMENT.md — ±2% for machine-measured levels, ±1% for hand-traced ones, a level mixing the two keeping the machine band — and Edge Review's conf score is a review-ordering number, never an uncertainty.
Run log	Timestamped record of the run: directory path, each landing, watchdog status, warnings.

SLDEA companion tools

Four stand-alone programs for recorded SLDEA runs — they touch no instruments, so any PC with a copy of the run data works.

🎚 EDGE TUNER

Drag sliders, watch the outlines redraw live on baseline / mid-run / late frames. 💾 **Save** writes the values into that run's `setup.txt`. Open it from the SLDEA tab (🎚 **Tune params...**) or double-click `Tune_SLDEA_Windows.bat` — or drag a run folder onto it.

🔍 EDGE REVIEW

Traces every frame of a run, queues the uncertain ones for a human pick, then writes the accepted areas back into the run's `data.csv`. Full walkthrough below.

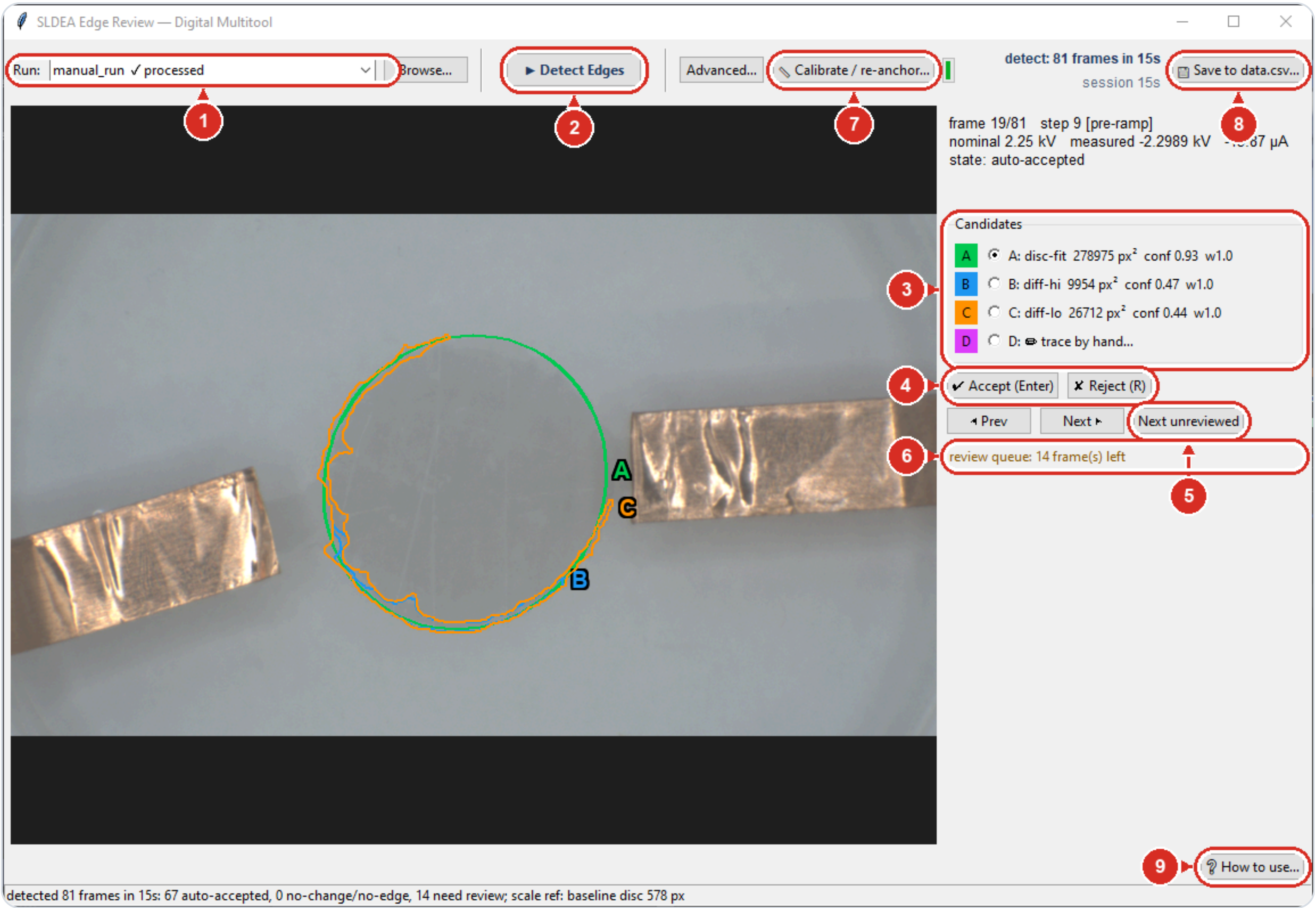
🩺 DIAGNOSTIC

When sliders can't fix a run, stop tuning and measure why: `Tune_SLDEA_Windows.bat /diag` writes `sldea_diag.txt/.json/.png` into the run folder and changes nothing. No window needed.

📊 PLOT TOOL

Puts several finished runs on one figure: `python sldea_plot.py RUN1 RUN2 --mode area`. Writes a 300 dpi PNG *and* the tidy CSV behind it, so any figure can be traced back to its numbers. `--mode current` works on raw runs, before anyone has reviewed a frame.

Edge Review — reviewing a run



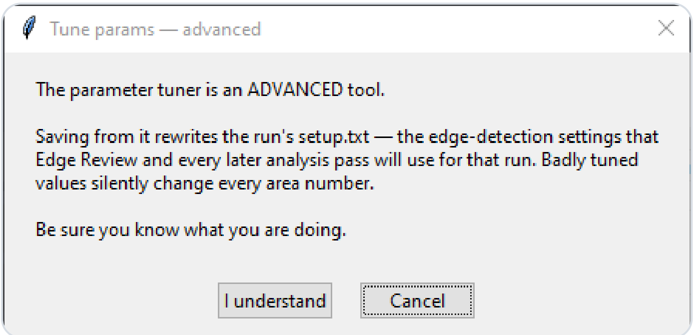
A real review frame (bench run P3_1 at 2.25 kV, pre-ramp): A = disc-fit (green), B/C = difference outlines. The machine's best pick is preselected — Enter agrees and moves on.

1. Pick the run — ✓ marks already-processed runs
2. Traces every frame — picks/rejects from earlier passes reset
3. Machine outlines — colors match the A/B/C tags on the image
4. Keys: 1/2/3 pick a candidate, Enter accepts, R rejects, D traces by hand
5. Jump to the next frame needing a human
6. Frames still waiting for a decision
7. Set or re-anchor the mm-per-px scale — verify the automatic disc fit
8. Writes accepted areas back (keeps a .bak) — confirms first
9. The review loop in one short read — start here

REVIEWING A RUN

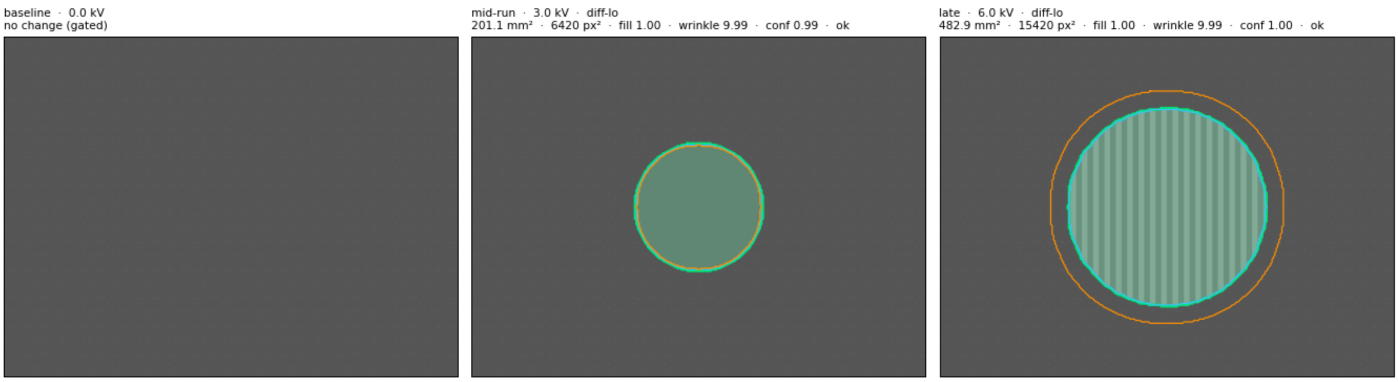
1. Open it from the SLDEA tab (🔍 Edge Review...) — or run `python sldea_edge_gui.py <run folder> --auto`.
2. Pick the run and click ▶ Detect Edges — a run with no scale anchor diverts to the 📏 dialog first: ✓ Accept the machine's disc fit (or measure by hand if it refuses) and detection continues. Most frames auto-accept; the review queue holds the uncertain ones.
3. On each queued frame, pick the outline that hugs the electrode: keys 1/2/3 (A/B/C), R rejects, 4/D opens the hand tracer.
4. Enter accepts and moves on; Next unreviewed jumps ahead.
5. If areas look wrong by a constant factor, 📏 Calibrate / re-anchor... — mid-review the corrected anchor is applied at Save; on a saved run it rewrites data.csv immediately, with no re-detection.
6. Queue empty → 💾 Save to data.csv... — read the counts in the confirm prompt before answering Yes.

When outlines look wrong — tuning and diagnosis

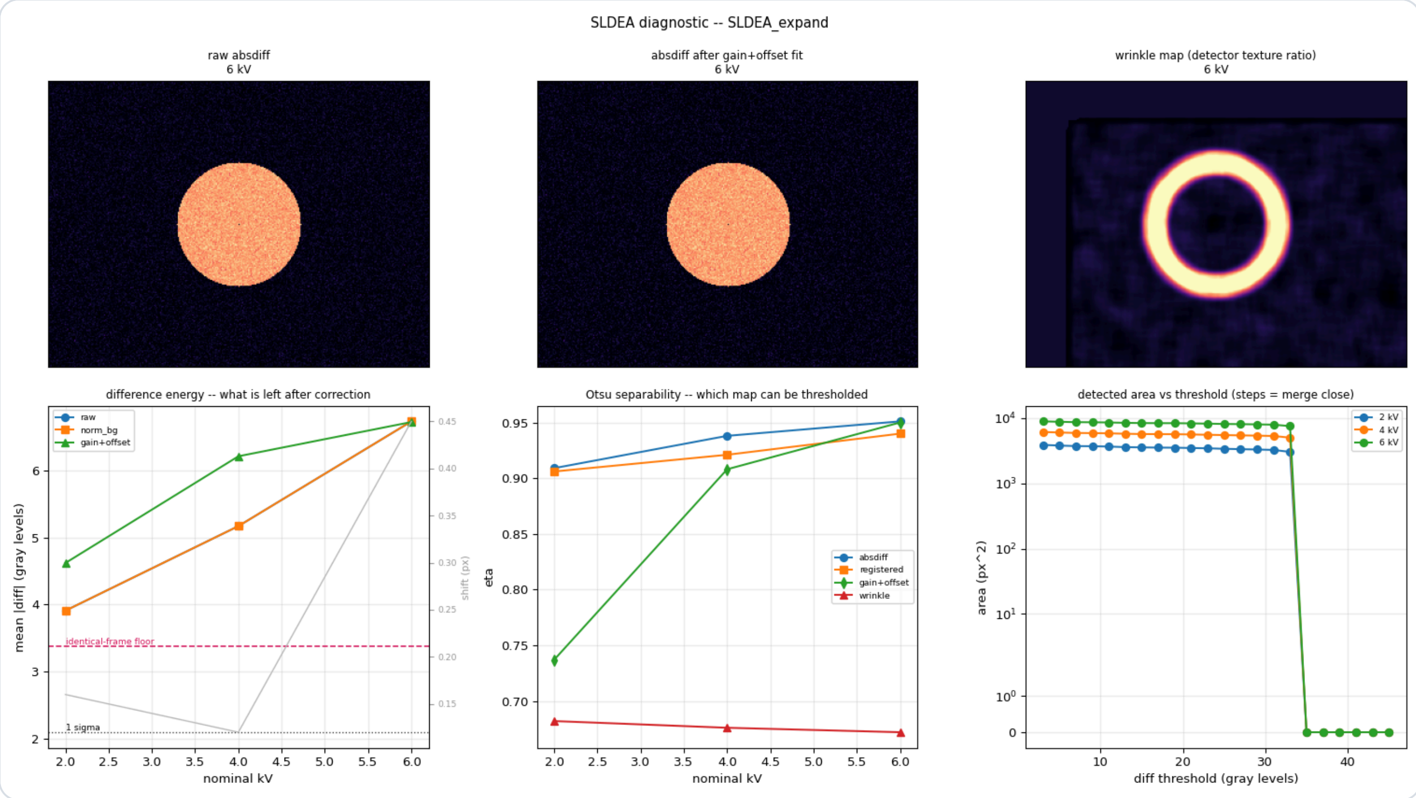


TUNE PARAMS IS AN ADVANCED TOOL

It asks you to confirm before it opens. Its **Save** rewrites the run's `setup.txt` — the detection settings Edge Review and every later analysis pass will use for that run — so badly tuned values silently change every area number. Reach for it only when Edge Review's outlines are consistently wrong across a run, and prefer per-frame picks or a hand trace for one-off bad frames.

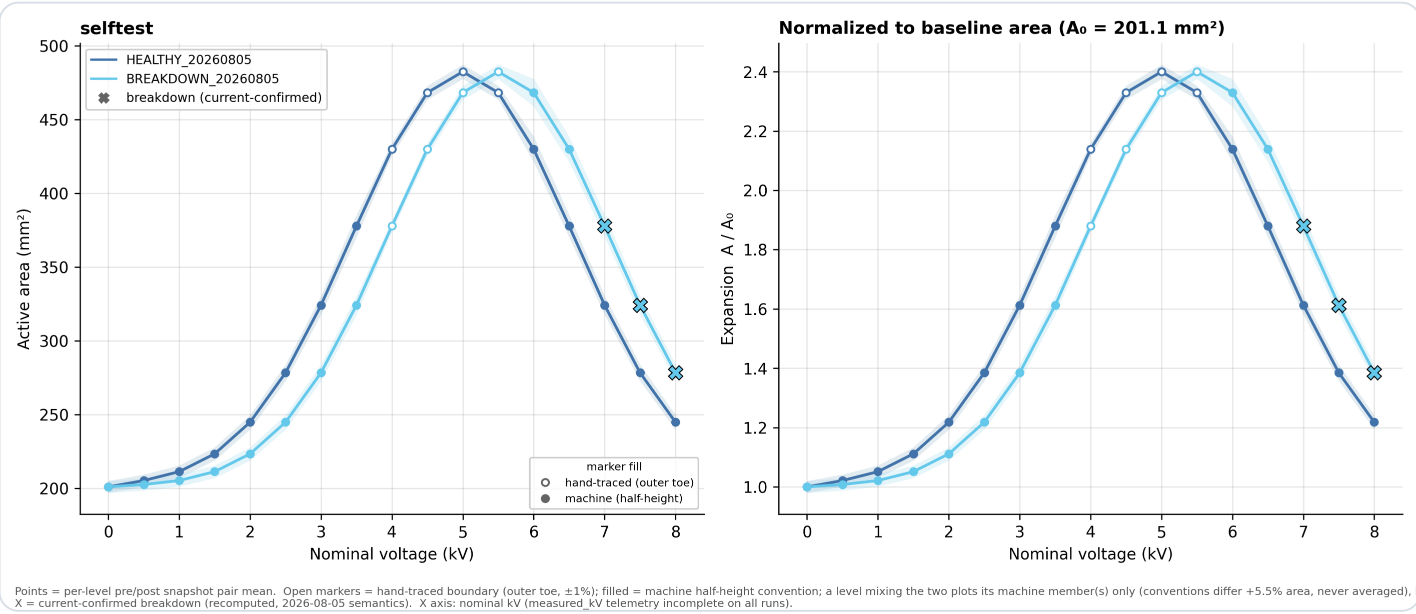


What the detector shows per frame: cyan = detected outline, orange = resting-disc reference. Confidence and area (mm^2) are printed above each panel.



The diagnostic's verdict panels — run `--selftest` any time to check the stack without data.

Comparing finished runs — the plot tool



The plot tool's own self-test figure. Left: active area against voltage. Right: the same curves normalised to each run's baseline, so runs that started at different sizes can be compared. Shaded bands are the measurement uncertainty; open markers mean a hand-traced boundary; ✕ marks a current-confirmed breakdown.

PUTTING RUNS ON ONE FIGURE

1. Review the runs first for `--mode area` — it plots measured areas, so it skips runs nobody has reviewed. `--mode current` and `--mode power` need no review at all.
2. `python sldea_plot.py RUN1 RUN2 --mode area --out ~/figures` — each RUN is a run folder, a parent full of runs, or a bench shortcut, exactly as for the tuner.
3. You get a PNG and a CSV of the same name. Keep them together: the CSV is the figure's evidence.
4. Breakdown ✕ marks are recomputed from the recorded currents every time, never read off the frame filenames — a saved brand the currents do not support is reported as stale rather than drawn.

TYPICAL TUNING SESSION

1. Copy the run folder to local disk (tuning re-reads frames constantly; the share is slow).
2. Double-click Tune_SLDEA_Windows.bat, or drag the run folder onto it (bench runs: just type 1, 2 or 3).
3. Check the panel labelled baseline really shows the 0 kV image.
4. Drag sliders until the outlines hug the active area in all three panels.
5. Click 💾 Save to setup.txt.
6. Open Edge Review on the same folder (--auto opens the scale-calibration dialog first — verify the machine's disc fit with ✓ Accept, or measure by hand if the fit refuses — and detection starts automatically once the anchor is in) and work the review queue.
7. 💾 Save to data.csv... when the queue is empty; if it still will not tune, run the /diag diagnostic instead of another slider session.
8. Once two or more runs are reviewed, put them on one figure: 📊 Plot runs... on the SLDEA tab (tick the runs, pick the mode, Export), or sldea_plot.py RUN1 RUN2 --mode area for the same figure from a script (--mode current works on raw runs, before any review).

WATCH OUT

> Tuner Save writes into that run folder's setup.txt. If you tuned a local copy, copy setup.txt back to the master run or the review pass will not use your tuning.

> If the run's baseline frame is missing or unreadable, the first loadable frame silently becomes the reference and every outline is wrong — always verify the baseline panel shows the 0 kV image.

> In Edge Review, ▶ Detect Edges starts clean: picks and rejects from a previous pass are discarded (hand-traced polygons are already safe in edge_labels.json). Save before re-detecting.

> Edge Review's Save rewrites data.csv (a .bak is kept) and renames frame files from the first breakdown onward with a _BREAKDOWN suffix — read the counts in the confirm prompt before answering Yes.

▼ Shortcuts, environment and every tool control

Tune_SLDEA_Windows.bat	Double-click, or drag a run folder onto it; builds its own small Python environment on first run. Add /diag to run the diagnostic instead of the tuner.
Bench shortcuts 1 / 2 / 3	Typing 1, 2 or 3 as the run argument opens one of the three bench runs; only live on PCs where the OneDrive folder is synced, inert elsewhere.
SCPI_SLDEA_DIR	Set once (setx SCPI_SLDEA_DIR "C:\SLDEA", new window required) and every no-argument launch — launcher, tuner, tab buttons — resolves the newest run under it.
Blur (px, odd)	Tuner slider: noise smoothing before differencing — higher is smoother.
Diff thresh (0=auto Otsu)	Tuner slider: fixed change threshold; 0 lets the algorithm pick one per frame.
No-change gate (diff p99)	Tuner slider: below this the frame counts as unchanged vs baseline (no outline).
Min fill / Search ROI / Electrode mask / Wrinkle-mode sliders	The remaining shape and mask settings; the grey hint beside each slider says what it does.
Match frame to baseline: gain+offset (norm_bg)	Cancels camera exposure drift before differencing; leave on unless comparing to an old run.
💾 Save to setup.txt / Reset to defaults	Save writes the tuned values into the run's setup.txt for Edge Review and auto-process; Reset restores stock values.
Run: + Browse... + ▶ Detect Edges (Edge Review)	Pick a run (✓ processed marks finished ones) and trace every frame. Beside the progress bar sit two clocks: detect: freezes when the detection pass ends and keeps its frame count (81 frames in 2m14s), so runs stay comparable, while session counts how long the window has been open. A run you only re-anchor the scale on never detects, so its detect: line stays at not run.
Candidates A/B/C + D: trace by hand (Edge Review)	Pick the correct outline per queued frame — keys 1/2/3 pick, R rejects, 4/D opens the hand tracer, Enter accepts and moves on.

📏 **Calibrate / re-anchor… (Edge Review)**	Sets the run's px→mm anchor. Required per run (the scale gate): Detect and Save are blocked until it is done, and it overrides every automatic scale reference. The dialog opens on the machine's automatic disc fit for you to verify — ✔ Accept adopts it; measuring by hand (circle or two-point) is the fallback, offered with the fitter's own reason when the fit refuses. On a saved run with px measurements and no open review pass the same button re-anchors instead: every mm² in data.csv is re-derived from the stored px and rewritten immediately, no re-detection needed — the dialog title says which of the two it is doing.
Advanced… (Edge Review)	Every detection setting with an explanation; Apply for this session, Apply + Save to write setup.txt. Re-run Detect afterwards.
💾 **Save to data.csv… (Edge Review)**	Writes accepted areas into the run's data.csv after a confirm prompt; keeps data.csv.bak and saves an area-vs-voltage plot plus outline overlays.
❓ **How to use… (Edge Review)**	Bottom-right of the window: the review loop in one short read — pick a run, Detect (calibration comes first, and the automatic disc fit is there to be verified, not repeated), work the queue, Save — plus the three things that cost people time: wash-out frames above ~5.5 kV are traced rather than rejected, an accepted scale is only staged until Save writes it, and a wrong scale can be re-anchored on a saved run without re-reviewing anything. It also says what conf and the w wrinkle index mean. Non-modal, so it can stay open beside the work.
sldea_diag.py (headless)	Run it on a run folder (or via /diag) when tuning fails; writes sldea_diag.txt/.json/.png into the run, changes nothing — the .json is small enough to send on.
sldea_plot.py (headless or windowed)	Puts one or more finished runs on a single figure. Takes the same run arguments as the tuner and the diagnostic (a run folder, a parent full of runs, or a bench shortcut) and writes a 300 dpi PNG plus the tidy per-snapshot CSV behind it, so a figure can always be traced back to its numbers. Colours are the house colourblind-safe set, assigned in the order runs are listed. Add --gui (or use 📊 Plot runs… on the SLDEA tab) for the window: the same figure, with the runs and the toggles as tick boxes and a live preview. The command line stays the way to script a batch of figures.

--mode area | current | power	area (default) needs reviewed runs and draws active area against voltage plus the same curves normalised to each run's own baseline; current and power work on RAW runs too, so a run can be looked at electrically before anyone has reviewed a single frame. Breakdown crosses are always RECOMPUTED from the saved currents, never read off the frame names — an old area-based brand once marked 35 healthy frames.

PART II — HOW IT WORKS

How it works — the instrument stack, PC to bench

Who this chapter is for

Part I tells you which button to press; this chapter tells you what is behind it — which cable each instrument really talks over, why the app is deliberately less capable on Windows, what an SLDEA run does minute by minute, and how code reaches the bench. Claims that came from an instrument rather than from reading source carry their date; anything only desk-tested says so.

Transports: which box speaks what

There is no single bus. The five instruments arrived over three transports, and every driver lives in instruments.py, one class per instrument. Three share a small base class that owns the PyVISA session, terminators and timeout; the DC supply and the DMM are stand-alone, because they are not USB-TMC devices at all.

Instrument	Transport	How it is addressed
LCR meter — BK 894	USB-TMC via PyVISA (pyvisa-py + libusb)	Auto-discovered by USB VID/PID. Its own USB serial descriptor is garbage, so the real serial only comes from *IDN?
Oscilloscope — MSO24	USB-TMC via PyVISA	Auto-discovered by VID/PID; the session is put into binary waveform transfer on open
Signal generator — BK 4055B	LAN first, USB-TMC as fallback	A fixed bench IP is probed first, USB discovery only if that fails. LAN is preferred because arb upload cannot work over USB at all
DC supply — BK 9174B	Plain serial over the built-in CP2102 USB-UART bridge	A fixed device node at 57600 baud, 8N1, CR+LF, from B&K's own example code. A virtual COM port, so it never shows up in a USB-TMC scan
DMM — BK 5493C	Raw TCP socket over LAN, opened through PyVISA as a SOCKET resource	A configured IP and a non-standard port. This unit's USB never enumerated and it is not VXI-11 either

Apply, then output — and read back what you can

The same convention runs through every tab that can energize something: Apply sends configuration, a separate button switches the output, and Apply never turns an output on. The drivers are built for it — the signal generator has a method whose only purpose is to set load and polarity without touching the on/off state, and the supply's apply() takes the output state as an optional argument defaulting to "leave it alone", because a supply may be powering something you care about. The output buttons confirm first, defaulting to No. The exception is the LCR meter's DC bias, which is part of the measurement configuration and so follows Apply like everything else on that tab — which is why loading a bench profile mid-measurement is called out as dangerous.

The GUI holds no authoritative model of instrument state: what it shows after a connect or a Reconnect was read out of the box, and the supply is read-only on connect so plugging in never changes an output. After Apply the app waits 200 ms and queries the channel back, deliberately best-effort — the configuration has already landed, so a slow query reports "read-back failed, use Read Instrument to refresh" rather than a configuration error. Where a value cannot be read the code says so instead of substituting a plausible number: a trace that has left the scope's visible window comes back as offscreen, not as the instrument's 9.9E37 sentinel dressed up as a reading.

Connect, Reconnect, and what closing the window does

There is no connect() method anywhere: constructing a driver opens the link, and a failed identify closes the session before re-raising — a leaked open claims the USB interface for the life of the process, and every later reconnect then fails until restart (audit 2026-07-25). At start-up the five are connected one at a time in a single worker, because the drivers share one process-wide PyVISA resource manager and it is not re-entrant. Reconnect runs the same code for one box — null the handle, close the old session, construct a fresh one — and is refused for the signal generator while an SLDEA run is live, since nulling that handle used to skip the run's safety ramp-down entirely.

- Closing the window stops a running SLDEA test first and waits up to about 3 s for its worker to zero the drive, then switches both generator channels off and turns LCR DC bias off.
- It hands every instrument back to LOCAL and closes the link, so the front panels work again without a power cycle — pyvisa-py asserts remote on connect and never releases it by itself.
- Closing the app does NOT switch off the BK 9174B DC supply output. Nothing on the close path touches it: it may be powering something, so it is left exactly as it was.
- Scope acquisition is not stopped either, and the generator keeps its configured waveform — only the output gate drops.

The 4055B's 52-byte USB ceiling

The generator's USB firmware (1.01.01.33R3) hard-wedges on any USB-TMC Bulk-OUT transfer spanning more than one 64-byte USB packet — even a whitespace-padded, pure-ASCII *IDN?. Once wedged, writes vanish, every read times out, and only a front-panel power cycle recovers it: INITIATE_CLEAR, *RST, a USB reset and a hub-port power cycle all fail, because the instrument is self-powered. Bench-verified 2026-07-02, three ways; it is also why EasyWaveX freezes the box, and the probe that demonstrates it lives in bench/archive/, labelled as intentionally destructive. Splitting a long command across several single-packet messages does not wedge the box but does not reassemble either — the first message is processed as the whole command and the rest dropped, so a 2 KB waveform upload stored as its first 24 data bytes, with the name registering, which is what makes that failure easy to miss.

So every USB command must fit 52 bytes including the newline: one 64-byte packet minus the 12-byte USB-TMC header. The driver enforces it rather than letting you find out — an oversized write raises with the byte count and the offending command, the basic-waveform setter auto-splits long parameter chains, and arb upload refuses over USB and points at LAN. Reads are unaffected. That is why arbs reach the instrument by LAN or by flash drive, and why the Arb Editor has two export buttons; the full story is in README §quirks.

The scope-monitor check, and the supply's dual range

The Trek's V_Out and I_Out monitors are ordinary BNC outputs into two scope channels (CH2 and CH3 by default), and nothing in that path is self-describing: with the scope set up wrongly a run still completes and simply records wrong numbers. Three real batches were lost that way — CH3 carrying a 10× probe factor on a plain BNC, inflating every logged current tenfold; CH2 at 2.6 mV/div for a 0–10 V monitor, so five runs recorded no usable measured kV (bench 2026-07-25); and a correct volts-per-division with the wrong window, clipping the 2026-07-29 batch from 4.25 kV up. So before a live run the app reads back each monitor's scale, external attenuation, position and offset, and checks three things: that the attenuation is 1× as a direct BNC requires, that the span covers the run's top voltage and trip current, and that the visible window actually contains that range. The check is polarity-aware, and the current window must contain ±the trip current, because every real breakdown in the 2026-08-04 ground-truth batch swung negative. A problem gives one dialog with three answers — fix and continue, run anyway, or stop — and fixing writes attenuation, scale, offset, position, coupling and channel-on for both monitors, then logs what it did. It is never silent, and a query that failed is reported as skipped, not as passed.

The DC supply has a smaller version of the same idea. Each of its two outputs is 105 W and dual-range: up to 3 A at or below 35 V, up to 1.5 A between 35 V and 70 V. The joint check lives in the driver's apply(), the only place that sees a voltage and a current together, and the GUI runs the same check on the main thread first, so an impossible combination is a readable dialog rather than a background traceback. One oddity when reading that driver: the 9174B has no channel-select command — channel 1 uses the bare SCPI token and channel 2 a 2-suffixed one (verified read-only against the unit 2026-07-22).

Windows edits, Linux drives

Instrument I/O works only on the Linux bench box. One module-level flag decides it, checked once at import; the reason sits underneath PyVISA, which drives the USB-TMC boxes through libusb with a udev and module-blacklist setup that exists only on that machine. On Windows the app still runs and the instrument tabs stay fully editable — what is locked out is connecting. The connection lines go amber with "Linux bench only -- view/edit (presets OK)" and Update Software... is greyed out. Those are normal on Windows, not faults: settings can still be typed, saved as presets and profiles, and loaded on the bench.

What works fully on Windows is most of the analysis work, not a token subset. Battery Data is not an instrument tab at all, just file processing; the Webcam tab takes snapshots and timelapses normally; and the whole SLDEA analysis chain — Edge Review, the tuner, the diagnostic and the plot tool — is stand-alone, touches no instrument and runs anywhere a copy of the run data does. The one Webcam feature that does not work off the bench is the sig-gen-driven stepped sweep, and it is worth knowing how it fails: the button is not greyed out but gated by a runtime check on the generator handle, which on Windows is permanently absent, so clicking it reports "Signal generator not connected". The effect matches the documentation; the mechanism is a null check, not a platform check.

An SLDEA run: the staircase and the two 0 kV frames

A run is one worker thread ticking at 10 Hz against a plan computed up front — voltage levels, ramp and hold segments, and the exact time of every photograph — which is why the tab shows the frame count and total duration before anything is energized. Start, End and Step become a list of levels; Up/down appends the descending half and Repeat multiplies the sequence. A leading 0 kV level is dropped from the landings rather than held, because 0 kV is captured as the baseline, not sat at. Each level contributes one ramp segment, one hold segment and two photographs, and the profile refuses to build if those timings collide with the landing time.

A run therefore takes two photographs at 0 kV. The first, tagged warmup, is thrown at the camera to make it settle and is used as a reference by nothing; the second, two seconds later and tagged baseline, is the frame every area in the run is measured against. It exists because of a measured failure: on the 2026-08-05 carbon-black run the baseline came out 73.7 % saturated while that same run's ramp frames were 0.27 %, because the camera firmware walks written control values back within about half a second. The tag matters more than it looks — warmup and not baseline-warmup on purpose, since the plot tool groups rows whose tag starts with "baseline" and a baseline-prefixed name would have averaged the throw-away frame into A_0, the exact number the warm-up protects. Every other consumer tests for the tag being exactly baseline. Both rows appear in data.csv; only one is the reference.

```
t = 0.0 s      frame  SLDEA_s00_00.00kV_warmup.png     tag warmup    (thrown away)
t = 2.0 s      frame  SLDEA_s00_00.00kV_baseline.png   tag baseline  <-- the reference
               ramp   0.00 -> 0.25 kV  over Ramp (s)
  + Settle     frame  SLDEA_s01_00.25kV_post-ramp.png  tag post-ramp
               hold   0.25 kV          for Landing (s)
  - Snap lead  frame  SLDEA_s01_00.25kV_pre-ramp.png   tag pre-ramp
               ...    one ramp + one hold + two frames per level ...
```

Control voltage, the Trek, and the monitors

The high voltage is not generated by anything in this rack. The signal generator is configured as a DC source and its offset is the Trek amplifier's control input, at 1 V per kV with a 10 kV ceiling; each tick computes the voltage the plan calls for and, only if it has changed, writes a new offset. That is the whole drive path. Coming back, V_Out reads 1 kV of Trek output per scope volt and I_Out reads 200 µA per scope volt, so 10 V on the scope is 2000 µA. Those constants, the gain and the ceiling live together at the top of sldea_profile.py — the only place the physical calibration is written down.

Trek inverts (negate control) is for the amplifier wired inverting, where a positive control gives a negative kV. Ticking it flips the sign of the commanded control voltage so the HV output stays positive, and sign-corrects the readings coming back — both monitors, not just V_Out, because the Trek inverts current and voltage alike (decided 2026-08-04). The pre-run window check is told about it too. The watchdog is the deliberate exception: it decides on the raw magnitude, before any sign correction. It is a drive setting rather than a description of the device, which is why it sits below the electrode and concentration fields rather than between them.

Snapshots, and the 2 Hz sidecar beside them

The photographs are the measurement: each writes a PNG into frames/ and appends one row to data.csv, flushed immediately so an interrupted run still leaves everything up to that point. telemetry.csv is a different thing — a continuous log of what the monitors did between photographs, at up to 2 Hz, with columns for elapsed time, timestamp, nominal and measured kV, measured µA, the two read statuses and an event field. The cap is 2 Hz because the watchdog already samples the current at exactly that rate, so the sidecar costs no extra instrument traffic; anything faster is a new duty cycle on the scope that nobody has timed. V_Out is sub-sampled to at most once a second, since the commanded voltage is already on every row, and the file is closed only after the HV is down.

Its honest status: the sidecar is desk-tested against a simulated scope and is NOT bench-verified. BENCH_TEST.md §M is its gate — a dry-run smoke test needing no HV — and until that passes nothing depends on the file. data.csv is still the run, and any rate figure quoted for the sidecar is an acceptance target, not a bench measurement.

The breakdown watchdog, and what "breakdown" means afterwards

The watchdog is armed only on live runs and only with a scope connected; ask for it without one and the run log says so rather than pretending. Before the staircase it learns the device's own quiet current at 0 kV — a settling pause, ten reads with the first two discarded, median taken — and refuses an implausibly large baseline, because a big "rest level" at 0 kV is a standing fault current and anchoring the trip to it would normalise the very signal the watchdog exists for; it falls back to the absolute rule, which then trips on the fault. From then on it reads I_Out every half second and trips only on sustained deviation from that baseline — by default 100 µA held for 3 s. A sample back below resets the clock, an unreadable sample is ignored without resetting it, and an off-screen reading counts as over-trip rather than as missing. If the current is unreadable for ten seconds the run raises a loud monitoring-lost warning and keeps going — the honest choice when the alternative is aborting on a flaky query.

On a confirmed trip the order is: log it, write the telemetry event, capture a frame tagged breakdown, stop the loop. The HV comes down through the same shutdown path every ended run uses, the run is marked BREAKDOWN-ABORT, and Edge Review is not auto-opened because that only happens for a completed run.

What counts as a breakdown afterwards is a separate and, since 2026-08-05, strict decision: only current-confirmed events rename frames with a _BREAKDOWN suffix. Confirmation means two adjacent rows deviating from the run's median current, or a terminal event. A collapse in measured area with no current signature is an advisory note and renames nothing. This is not fussiness — the old area-jump heuristic branded 35 healthy frames on run P3_5 while the measured current never left ±11 µA of its baseline. Renaming happens in Edge Review, never during the run, and it is reversible: un-branding is symmetric, and files are renamed, never deleted.

What lands in the run folder, and when

setup.txt is written first, before the camera is even configured, so an interrupted run still leaves a readable record of what was attempted. It is a lab-notebook document, not a config file, and it stays plain text — decided 2026-08-08; if typed or nested structure is ever genuinely needed it arrives as a sidecar beside the txt rather than by converting it.

File	Written when	What it is
setup.txt	At run start, before camera setup; appended by Edge Review later	The lab notebook: drive settings, monitor scaling read back off the scope, camera, device (diameter, electrode, concentration), the snapshot plan
run.log	From the moment Run is pressed	Everything the Run log panel shows, including the pre-run checks
data.csv	Header at start; one row per photograph, flushed immediately	The run. The area columns are left blank at capture time and filled by the edge-detection pass later
frames/*.png	One per snapshot	SLDEA_sNN_XX.XXkV_tag.png — step, nominal kV and tag. Renamed with a _BREAKDOWN suffix only later, and only by Edge Review
telemetry.csv	Continuously at up to 2 Hz; closed after the HV is down	The sidecar. Nothing reads it to measure anything
data.csv.bak	Not by the run — only when Edge Review saves	One backup copy, overwritten on each save

Stopping a run: ■ Abort, and the window-close path

■ Abort sets a stop flag. The loop checks it at the top of every 10 Hz tick, so it exits within about a tenth of a second, and the shutdown runs in the worker's finally block: the control voltage is commanded to 0 V and only then is the output opened. The safety property is that ORDER, not a slew rate — what the code does is one write of the offset followed by the output switch. It tries the generator handle the run captured at the start and, if that is no longer the attached one, the current handle too, so a mid-run Reconnect cannot orphan the instrument that was actually driving. If neither can be zeroed the run does not end quietly: a loud line goes into the run log and a modal HV NOT ZEROED dialog tells you to switch off the generator and Trek at the front panel. An earlier version swallowed that failure and ended the run green with the Trek still energized.

Closing the window during a live run is a weaker path. It sets the same flag and waits up to about 3 seconds for the worker to finish — but it is best-effort. If the worker has not reached its finally block by then, close falls through and cuts both generator outputs bluntly while the offset may still be at full scale; and because the window is then destroyed, the HV NOT ZEROED dialog can never be shown. That is the whole reason the documentation insists on it: a live SLDEA run gets only a best-effort ramp on window close — ■ Abort is the real stop.

From main HEAD to the bench: the deploy chain

Nobody runs this from a git checkout on the bench. The Linux desktop icon calls a wrapper that tries, in order: the ShareDrive copy, a GitHub fallback clone on local disk, this user's local cache, then a developer clone — and if all four fail it shows a visible error rather than nothing. The share is probed by reading a byte from it, not by testing readability: on 2026-07-20 a readability test succeeded against a dead NAS and the launch died with "Host is down". The fallback clones to local disk deliberately, because git on the CIFS share corrupts packfiles. On Windows the launcher does the same job in five printed phases — check the share, find a Python 3.10+ with tkinter, mirror the app locally, build or update a virtual environment, launch — with the working directory left on the share so presets stay shared.

The updater that refreshes the share does a shallow clone of main and rsyncs it across. It passes no branch and no tag and never touches the GitHub releases API, so what the fleet gets is main HEAD, whatever that is. The consequence is worth internalising: publishing a release as a pre-release does not hold it back — a pre-release reaches every bench the moment its commit is on main. Tags govern which PDF the releases page serves; they do not govern the code anyone runs.

The version stamp is what makes redeploying safe to repeat. version.py holds the semantic version and version_string() appends the short git hash at runtime when it can find a checkout; the share copy has no .git, so the deploy script bakes the hash into the version string instead — last, after everything else has synced, because stamping first once let a launch mid-deploy cache half-copied libraries under the new stamp permanently (audit 2026-07-25). Both launchers read exactly that one file to decide whether their cache is stale. The stamp is a deploy-complete marker, not a release identifier — which is precisely why a pre-release propagates.

If you are extending any of this: new SCPI paths do not ship without a bench session, and anything touching an HV-safety path — the runner, the watchdog, instrument output switching — gets an adversarial review pass before the pull request opens.

PART II — HOW IT WORKS

How it works — Edge Review, from photo to mm²

Who this chapter is for

This is for the labmate who wants to understand the measurement, or extend it — not to drive it. Driving it is the Companion tools chapter and Edge Review's ❓ How to use… panel, which lives inside the app so it ships with the code it describes.

Every number here is measured and carries its date. Where a term is genuinely unmeasured — and one important one is — this chapter says so rather than rounding it to a reassuring figure.

What a run is, on disk

A run is a folder the SLDEA tab wrote; Edge Review never invents one. A typical run is ~40 voltage steps to 10 kV with two snapshots per step — pre-ramp and post-ramp, about 57 s apart — so roughly 81 rows at 1920×1080.

- setup.txt is a lab-notebook document, not a config file, and it stays plain text (decision 2026-08-08). Machine-read lines keep the Key: value convention through se.load_settings / se.save_settings; if typed or nested structure is ever needed it arrives as a sidecar beside the txt, never by converting the document.
- Two machine-read blocks sit inside it: the edge-settings section, and the scale-anchor block. The anchor block comes FIRST so replacing the settings cannot delete it, and every key in it is optional on read — 15 runs carry the 2026-08-05 two-click anchor without the newer fields and two predate the block entirely. Both files are read UTF-8 with errors='replace': one hand-annotated non-ASCII byte used to raise mid-dialog and silently brick Detect for the session.
- telemetry.csv is a sidecar nothing in the analysis chain reads: run_csv matches ^data(\d*)\.csv$, so it can never be mistaken for the run CSV.

```
SLDEA_<timestamp>/
    setup.txt                   lab-notebook document + two machine-read blocks
    data.csv                    one row per snapshot
    frames/SLDEA_s01_00.25kV_pre-ramp.png …
    telemetry.csv               live runs since 2026-08-05 (watchdog log, ~2 Hz)

    edge_labels.json            hand traces, append-only ground truth
    scale_calibration_log.txt   one line per completed calibration round-set
    overlays/                   accepted outline drawn on each frame   } written
    area_vs_voltage.png         the session's accepted areas           } at Save
    data.csv.bak                pre-save backup
```

data.csv, and which column is primary

- Because px is primary and mm² derived, a wrong scale is recoverable and a wrong outline is not. That asymmetry is why the re-anchor path exists.
- Never use an active_area_mm2 written before 2026-07-28: the old blob detector's scale bug understated them by 2.3–2.7× in area. Reprocess; do not mix the two eras.

Column	Written by	Meaning
snapshot, step, tag	capture	Index, staircase step, and which of the pair: baseline, pre-ramp, post-ramp, or warmup (the throw-away frame at t=0, runs since 2026-08-05).
nominal_kV, control_V	capture	Commanded voltage and the control level behind it. The X axis of every figure — measured_kV is incomplete on every run in the 2026-07/08 batch.
measured_kV, measured_uA	capture	Scope readings at the snapshot, Trek-polarity corrected. Blank when unusable.
frame_file	capture; rewritten at Save	The PNG this row measures. Rewritten only for breakdown branding, and only to a name that exists or is about to.
active_area_px	Save	The accepted outline's area in full-resolution pixels. The PRIMITIVE measurement — every mm² and every A/A_0 derives from it.
active_area_mm2, active_diam_mm	Save	The same measurement through this save's px→mm anchor. Derived, never primary.
wrinkle_idx	Save	The wrinkle index w of the accepted outline, measured on the normalized frame.
notes	Save	Method and confidence (edge:disc-fit conf 0.94), plus flags, advisories and consistency annotations.

What Save commits, and what a trace banks immediately

- Nothing from a review pass reaches data.csv until 💾 Save. The one exception is a re-anchor on an already-saved run, which says so on its own button before it writes. Save is also gated on the scale anchor: with none recorded it refuses, because no entry point may write mm² off an unverified scale.
- Save's destructive phase is ORDERED — columns filled, rename plan computed, data.csv committed (backup, then atomic tmp+replace), and only then are frames renamed. A failure before the commit leaves the run byte-identical with zero renames; the old order renamed first, so a full disk left renamed frames, a stale CSV, and a dialog promising a .bak that was never written.
- ONE SCALE PER SAVE. A row still in the review queue keeps its previous pass's active_area_px, but its mm² and diameter are RE-DERIVED at this save's scale. A partial re-save used to write two absolute scales into one column — a 56.1 % artificial step on the real-data repro, invisible to the breakdown rule and the plot tool because both read px. A stale mm² with no px is blanked rather than kept on an unknowable anchor.
- Tracing banks separately and immediately: the polygon is STAGED as candidate D and Accept commits it, but the label is appended to edge_labels.json at DONE, not at Accept — a completed trace is ground truth whether or not it is committed, and a re-trace appends another record, which is how operator repeatability was measured. Tracing never touches data.csv or setup.txt, and the sidecar is append-only, written tmp+replace, refusing rather than clobbering a corrupt file.

How a frame becomes candidates

Detection is a competition between channels refereed by an acceptance system; up to three candidates survive per frame, best first. First the frame is mapped into the baseline's photometric space by a gain+offset fit computed on PAPER ONLY — ROI minus disc minus electrode footprint — so a changed device cannot drag its own correction. The footprint is defined by Laplacian-energy texture rather than brightness, because most of the copper is dimmer than the paper (median foil pixel 173 against paper 176–190).

The scene sets the channel order. The resting disc sits only 10–25 gray levels below the paper (162–167 against 176–190) while the strips span 120–255 across 12.7–13.3 % of the frame: the device is the lowest-contrast object in its own photograph. And it activates by WRINKLING with almost no intensity displacement — separability of the plain intensity difference measured 0.000 on all 72 frames of the P3 campaign, against ~0.3 for the texture map. Thresholding a difference image is therefore not the primary signal here, which is why the diagnostic's [HIGH] verdicts about it ('The difference is dominated by photometry, not by the device'; 'No map here separates at all'; 'One threshold cannot serve this run') fire on healthy, reviewed, PASSING P3 runs. They are true statements about a channel the area does not come from; on the same run the verdict about the measurement reads OK — 'Accepted boundaries sit on a real ink step'.

- The honest no-change gate: below min_diff (default 6.0) on the ROI difference's 99th percentile, the intensity tiers return nothing — inventing an outline there was the old failure mode. Texture and boundary are consulted either way, because a fine wrinkle can be invisible to the downscaled difference and still be real. With no readable baseline at all, candidates() returns nothing: the old base-less fallback thresholded the raw photograph and auto-accepted the ROI background at conf 0.85–0.90, 2.74× the true area.
- disc-fit: 360 rays from the resting centre; per ray the strongest sustained dark→light ink step, searched between 0.80 and 1.38 r_0, with a cut adapting to the scene's own median ink contrast (max(3.0, 0.35 × median)) because P3 ink steps 10–25 gray levels while the 07-23 devices step 40+; sub-pixel parabolic refinement; strips and their leads excluded by azimuth (±10°); robust trimmed ellipse; area from the FITTED shape. It refuses unless the fit lands between 0.9 and 1.3 r_0, holds its centre within 0.15 r_0, reaches circularity ≥ 0.55 and a median residual ≤ 6 % of the equivalent radius, on ≥ 40 edge points from ≥ 90 open sectors.
- The ink edge is the only feature that IS the boundary, and the two alternatives were built and falsified on the frames: a change-map boundary rides out to the passive membrane ring that hoop-wrinkles around the disc (1.6–2× areas), and a valley tracker follows the taut rim, which migrates INWARD with kV so the area shrinks as voltage rises. At 4.25 kV the ink step visibly moved ~80 px and the fit sits on it.
- The wrinkle index w is a texture ratio: blur frame and baseline (σ 2.5), take mean |Laplacian| inside the outline in each, divide frame by baseline. 1.0 = no change, clipped at 9.99. The pre-blur is load-bearing — raw Laplacian is dominated by sensor grain and read ~0.9 on obviously wrinkled frames. wrinkle_ratio does two jobs: the tex-ratio segmentation cut, and the threshold at which a row is annotated wrinkle-mode (informational only — it never triggers breakdown renaming). Hand traces compute w on the NORMALIZED frame exactly as the detector does; |Laplacian| is linear in gain, so the old raw-frame path put values 20–30 % off into the same column as machine ones.

Channel	What it actually measures	Standing
disc-fit	The ink edge of the known disc, ray-tracked and fitted with a robust trimmed ellipse — the full responding disc, leads and passive wrinkle ring excluded.	Primary. Median IoU 0.89 vs operator traces (n=26, min 0.82).

resting	A statement, not a detection: on a frame with no change against the baseline, the area EQUALS the resting area.	The low ramp. Confidence grows with the margin below the gate, capped 0.95.
tex-ratio	The wrinkled REGION — local Laplacian energy of frame over baseline, cut at wrinkle_ratio (1.4).	Where the fit refuses. Region, not boundary: IoU ~0.43, area −40..−69 %.
diff-lo / diff-otsu / diff-hi	The changed region at 0.6×Otsu, Otsu and 1.5×Otsu of the ROI difference.	Secondaries. Same region-not-boundary caveat.
manual-trace (candidate D)	The operator's polygon.	Ground truth by definition; the only measurement above ~5.5 kV.

From candidates to an accepted row

Ranking and acceptance are separate machines, and keeping them separate is load-bearing. Ranking decides which outline is the best measurement on offer; acceptance decides whether a human must look at it. A candidate can keep first place and its area while losing the right to be accepted unseen.

- Containment cap: a tex-ratio or diff patch whose centroid is inside a valid disc-fit and whose area is ≤ 1.2× the fit's is supporting evidence, not a better answer — capped just below the fit. A region LARGER than 1.2× is a disagreement and still competes. Extended from tex-only to the diff tiers on 2026-07-29 after an operator spot-read: on 155425 at 5.25 kV a diff-hi patch at half the fit's area had outranked the correct boundary.
- Ramp hysteresis: the previous frame's winning channel gets +0.05, so a challenger must win by a margin rather than a coin flip — near-tied channels flipping on single frames caused most same-kV pair mismatches. The bonus is applied BEFORE the sort, so it can change which candidate ranks first, not merely its score.
- Same-kV pair reconciliation: the two snapshots of one step are independent detections of one physical state. Agreement within a tolerance derived from their own fit CIs gives both +0.05; disagreement past twice that tolerance caps both just below accept_conf, so a confident tier flip cannot auto-accept on both sides of a contradiction.
- Auto-accept then takes any frame whose best candidate has conf ≥ accept_conf (0.75), spread within spread_pct (12 %), and no fallback tag. A frame with no candidates is auto-rejected as an honest no-change/no-edge; an UNREADABLE frame is neither, staying in the queue labelled with its true cause with its saved values intact — before that distinction existed, an undecodable file was laundered into 'no change vs baseline (auto)' and Save blanked a hand-traced terminal-breakdown measurement.
- accept_conf = 0.75 is FINAL for the current label set (2026-07-29 at 34 labels, unchanged at 47). Raising it buys nothing — wrong high-confidence patches are already held by the spread and pair rules — and lowering it was rejected on area evidence: the audit-capped onset fits it would recover run a median −6.9 % area against the operator.

conf is a review-ordering score, not an uncertainty

conf measures the strength of internally consistent evidence. It is valid for ORDERING frames for review. It is not a calibrated probability that the boundary is correct, and it is not an error bar: read conf ≥ 0.85 as 'no internal contradiction found', never as 'validated correct'.

That is measured, not cautious. The first calibration round (13 operator traces on 155425, 2026-07-29) found conf ANTI-calibrated: the 0.50–0.75 bin scored P(IoU ≥ 0.8) = 0.40 while both bins above it scored 0.00 — every high-confidence label was a diff-hi winner at 5.5–6 kV outlining the interior wrinkle patch, pair-confirmed to 0.97–0.99 because both snapshots were fooled identically. Pooled at 47 labels, method identity predicts correctness and conf does not: disc-fit median IoU 0.89 (n=26), patch tiers 0.43 (n=15) with recorded areas −40..−69 %. The real uncertainties live in the error budget instead.

- The per-frame random term is on the candidate itself: disc-fit's spread_pct is its own 85 % confidence interval on area from the edge-point scatter — 0.2–0.5 % on the P3 runs, 0.5–0.7 % on the 07-23 runs. It replaced a cross-tier spread that measured threshold sensitivity, a different quantity.
- The largest single term is definitional, not statistical. On audit-clean control frames the operator's trace sits +5.2–5.7 % in area above the machine's, with a spread across four independent controls of only 0.5 %: the machine takes the HALF-HEIGHT of the ink step, a human tracing by eye lands on the OUTER TOE. Pick a convention, state it, and absolute areas inherit only the scale terms and the per-frame CI.
- Ratios are tighter for a geometric reason: the offset behaves as a near-constant annulus of ~7 px on a ~289 px resting radius, and a fixed annulus cancels to second order — at 1.44× area the residual between conventions is ~0.8 %. Honest caveat: it was measured at rest and low kV, and the onset-frame excess of ~1–2 % bounds how much it could grow under large strain.
- No machine boundary should be judged against a bar above IoU ~0.97: intra-operator repeatability measured median IoU 0.973 over 9 repeat pairs, area deltas 0.2–2.5 %.

Quantity reported	Quote	Dominated by
Expansion ratio A/A_0	±1–2 %	Fit CI plus the second-order residual of the edge-definition offset — the scale cancels exactly.
Absolute mm², convention stated	±1–2 %	Scale anchor plus fit CI.
Absolute mm², convention NOT stated	±3 % (or a one-sided +5.5 % band)	The edge-definition offset. Avoid this — state the convention.
Hand-traced areas (wash-out ≥ 5.5 kV)	±1 % precision, outer-toe convention	Operator repeatability. The machine has no boundary there; the traces ARE the measurement.

The scale chain: one anchor per run

Everything absolute hangs on one number — how many millimetres a pixel is worth. Three links: a nominal diameter, a measurement of the resting disc in pixels, and a decision to adopt it. The nominal diameter is CLOSED: the CNT electrodes are applied through a laser-cut mask, so the resting discs sit at 16 mm by manufacture for this whole series (lab confirmation, 2026-08-01) — a machined constraint, common-mode across every device cut by it.

The pixel measurement is the machine's job, and that is a measured conclusion rather than a preference. On 2026-08-06 an operator ran eleven interleaved hand calibrations on one disc against an automatic fit of 577.08 px: per-fit precision came out at σ ≈ 1.0–1.1 % of diameter and did NOT move with method or stroke (circle at 3 px 1.03 %, the same circle at 1 px dashed 1.11 %, two-point-with-rotation 2.09 %), and the fit beat all eleven on accuracy and nine of eleven on precision. The mechanism, measured on the frame: the disc reads 166 gray, the paper 186, and that 20-level step is spread over ~60 px of radius. There is no line to click, and the point a human picks is the outer toe — the 3 px stroke's measured +2.07 % diameter bias. Averaging suppresses noise as √n and does nothing to a bias, so no round count would have fixed it.

The corpus agrees. A sweep of every baseline (2026-08-06) fitted 13 of 15 runs at residuals of 0.03–0.80 % of diameter, and each of the eleven recorded resting areas is predicted to two decimal places by its anchor's deviation from that fit; the eight runs with no manual anchor land on $\pi \cdot 8^2$ exactly. Manual calibration is the only source of absolute-area error in this dataset. So the scale gate opens on the machine's measurement and asks the operator to VERIFY it; hand measurement is the fallback for when the fit refuses.

- baseline_disc refuses rather than fabricating: 360 radial rays from a central dark seed, the strongest dark→light step per ray where the neighborhood is clear of foil and glint, a robust circle fit run twice (an off-centre seed truncates the far side) — and nothing at all unless the accepted edge covers ≥ 120° of arc, the residual is under 6 % of the radius, the circle is filled with the dark class, the diameter is a plausible fraction of the search window, and circularity is ≥ 0.85. Its predecessor returned an 85 %-of-frame blob at circ 0.32 and conf 0.93, and every recorded mm^2 inherited a 2.3–2.7× area error from it. On a refusal the dialog quotes the fitter's own reason from eleven individually named gates, because 'the arc covers only 87°' and 'the diameter is outside the plausible range' send the operator to different places.
- Both signed-off methods outrank every automatic reference at Save: an approved fit must keep beating an accepted result on the baseline row exactly as a hand measurement does, or a baseline-row detection would silently outrank the number the operator signed.
- Verify-then-Accept STAGES the anchor for the next Save. The one path that writes immediately is the RE-ANCHOR on an already-saved run — every recorded mm^2 re-derived from the pixel measurements already in data.csv, with no detection and nothing re-reviewed, because only the scale was ever wrong. It is not new code: the commit re-runs the normal fill with an EMPTY results dict, so every row takes the unreviewed branch. It refuses when no row carries px, when a detect worker is in flight, and when an UNSAVED review is in memory — the last being the mixed-scale bug's own shape.
- Verified on scratch copies of three runs against the real frames: fits reproduced at 577.08 / 370.65 / 362.18 px and each resting area moved to 201.06 mm^2 = $\pi \cdot 8^2$. That landing is an arithmetic identity — declaring the fitted disc to be 16 mm forces it — so a resting area that does NOT land there means the anchor is not the fit. A/A_0 recomputed from active_area_px is BIT-IDENTICAL across a re-anchor; from the stored 3-decimal mm^2 it agrees only to ~1e-5 relative (worst 1.2e-5). area_vs_voltage.png goes stale — it is drawn from a review pass's accepted results and a re-anchor has none — and is named rather than deleted; the overlays are px contours and stay correct.
- Calibration methods are recorded as NAMES — verify, circle, twopoint — never dialog letters. The letters were renumbered on 2026-08-06 and the old ones are on disk, so a stored LETTER means its pre-swap meaning (A = circle, B = twopoint, C = verify), applied in one place, se.cal_mode_read. Live data contains both vocabularies.

Recorded method	σ / SE	Independent cross-check?
auto-verified — the operator was shown the automatic fit with its quality numbers and approved it	UNDEFINED: one fit, no rounds. Never written as 0.	None exists — see the next section.
manual-calibration — the operator measured the disc themselves, n blind rounds, averaged	$\sigma \approx \mathrm{range}/d_2(n)$; mean SE = $\sigma/\sqrt{n}$; area SE = 2×SE.	Yes: the automatic fit, and the 16 mm mask's $\pi \cdot 8^2$ resting area, both gated at ~1 %.
(no anchor block) — a pre-gate run	—	—

The honest gap in the scale chain

An auto-verified anchor has no independent cross-check, and the code says so rather than faking one. Declaring the fitted disc to be diam_mm makes the resting area $\pi \cdot (\text{diam\_mm}/2)^2$ BY CONSTRUCTION, so the anchor guard's two tests both return exactly +0.00 % on such an anchor — on any frame, at any diameter, however wrong the fit is (pinned over a 12→900 px sweep). A green tick from a check that cannot fail is worse than no check: it reads as verification the code never performed. So the guard is not run on a verify-mode anchor and is not claimed anywhere — not in the dialog, not on either status line, not in the run's record, not in the diagnostic. All five surfaces say in words that no cross-check was performed and that no independent one exists. The verification is the operator's eye, supported by the fit's own quality numbers.

What DOES quantify such an anchor is the fit's median edge-point residual as a fraction of diameter: on P3_2's baseline, 2.3 px on 577.08 px = 0.40 %, which lands on the standing budget. It is deliberately conservative — per-point scatter, not the standard error of the fitted radius, which with 204 edge points is roughly $\sqrt{n} \approx 14\times$ smaller (~0.03 %). And it is a PRECISION figure. The fit's SYSTEMATIC term — which feature the step-finder locks onto versus the true mechanical boundary — is measured by NOTHING in this project, and cannot be measured by re-fitting: a 0.03 % residual says the edge points lie on a circle, not that it is the right circle. The only external evidence is baseline_disc agreeing with by-eye readings to ~1 % on three P3 baselines, plus the eleven-attempt comparison above.

- The hand-calibration σ figures are likewise not a distribution: one operator, one disc, one session. They are quotable as exactly that, and they justify the verify mode. Until several runs and both hand methods have been measured, SLDEA_MEASUREMENT.md §2.1's ~0.4 % diameter / ~0.8 % area remain the numbers to quote, and the two-point mode's own σ is entirely unmeasured.
- For a hand anchor the gate is the MEAN's standard error, not the raw range: SE ≤ 0.4 % of diameter (≡ 0.8 % area), derived from the standing budget rather than invented. A range cannot shrink when a round is added, so a range gate's remedy could never clear it; SE falls as $1/\sqrt{n}$, so the gate can name the round count that would. The range→σ conversion is n-aware (control-chart d_2 factors, n = 2–8) and REFUSES an n it has no factor for, returning σ and SE undefined together rather than reusing the n = 3 constant. Even here the two guard references are not fully independent: baseline_disc returns a circle, so its area is πr^2 and the area test is the diameter test squared.

The two audits

Confidence certifies consistency; the boundary self-audit is the correctness check, and it costs the operator nothing. For every winning disc-fit or resting candidate, 180 rays are cast along the candidate's own contour (per-angle radius, so an ellipse is not penalized). Each open ray reports the SIGNED offset between the fitted boundary and the measured half-height of the ink step under it (+ = the fit sits outside the step), and whether there is a measurable step there at all; rays through the foil are skipped, not repaired. 'Circled the noise' reads as a large no-step arc, a systematically wrong FEATURE as a nonzero bias with a small MAD.

The audit gates ACCEPTANCE, not ranking: a tripped winner keeps its rank and its area — still the best measurement on offer — and loses only the right to auto-accept, tagged so the queue says why. Pair agreement can never lift it back, because two snapshots interpolated over the same washed-out arc agree beautifully, and that is exactly the correlated error pair agreement cannot certify against.

- The bias gate found something nobody was looking for. On all six bench runs, in the 1.5–3 kV band, frames were winning as resting at conf 0.84–0.99 and auto-accepting while the audit measured the ink step OUTSIDE the claimed circle — bias −4 px at 1.5–2.2 kV growing monotonically to −11 px (P3_2 at 2.0 kV), understating the area by up to ~7 %. The audit had seen it all along; nothing was listening.

- The resting-refit measures rather than asserts: when the bias gate trips on a resting winner, the fitter runs on that frame with the change-map 'is it responding' gates WAIVED — the audit has just proved a measurable step exists off the circle — while every ink-profile quality gate still applies. The refit is audited like any winner and takes the frame only on its own merits; the capped resting claim stays as runner-up. Validated against stored labels: P3_1's 2.0 kV pair refit +4.6/+4.1 % where the audit predicted +3.8/+4.2 %.
- Read the per-frame audit_nostep and audit_bias fields as EXCEPTION FLAGS, not measurements. They are set only when a frame EXCEEDS its gate, so null means the frame PASSED — not that the check was skipped. The real per-frame numbers live in frames[].audit in the diagnostic's JSON. Reading the flags as measurements turns a pass into 'unevaluated', which has been done once and corrected.
- Percentiles follow np.percentile's LINEAR convention, matching the diagnostic's own report and every run's sldea_diag.txt. Nearest-rank differs on six of twelve runs in the corpus, so do not 'correct' one method toward the other.
- At run level the verdict stands: median audit bias stays within −0.4..+0.4 px on all six runs, bounding wrong-feature tracking below ~0.3 % of area — the caps are per-frame exceptions, not a systematic. One run needed its own knob: 155425 has a stable ~15.5–16 % faint-arc sector on every low-kV frame (a benign floor, bias clean there) sitting just above the 15 % default, so audit_nostep_pct: 20 is saved in that run's setup.txt. Do NOT raise the default — P3's floor is 0–2.8 % and the default catches real onset interpolation there.

Gate	Setting	Trips when	What it usually means
audit_nostep	audit_nostep_pct, default 15 %	More than 15 % of the fitted arc has no measurable ink step under it.	Wrinkle onset: the ink locally washes out and the ellipse INTERPOLATES those sectors.
audit_bias	audit_bias_px, default 3 px	The median signed offset between fitted boundary and measured step exceeds 3 px.	A stale 'resting' claim — the disc crept out below the no-change gate — or a fit sitting off the ink.

Breakdown: what renames a frame

Since 2026-08-05 only CURRENT-CONFIRMED events rename frames with a _BREAKDOWN suffix; area collapse alone is an advisory note and nothing else.

The rule is deviation-based. The baseline is the per-run MEDIAN of every parseable measured_uA — robust even at the worst observed 31 % event contamination — and an event row deviates from it by at least breakdown_dev_ua (20 µA). CONFIRMED needs two event rows on ADJACENT CSV rows, or the run's last parseable-current row being an event (terminal, no recovery evidence). A single event followed by recovery is a self-clearing transient: advisory only. A blank-current row between two event rows BREAKS adjacency, deliberately — at ~30 s between snapshots there is no evidence the excursion spanned the gap. Absolute thresholds cannot do this job: the whole 07-29 campaign sits on a stiff −16 µA instrument offset present at the 0 kV baseline row, so abs(µA) > 50 was really 34 µA of headroom one way and 66 the other. The absolute rule survives only as the fallback when a run has fewer than five parseable current rows.

- The 2026-08-04 ground-truth table set the threshold: true events sustain 26.5–208 µA of deviation, false ones stay at or below 14.6, recovered transients reach 48–137 before returning. Every value between 14.6 and 26.5 separates the corpus; 20 splits the margin.
- The incident behind the rule: P3_5 recorded 'breakdown? area collapsed 36 %' at 5.75 kV and renamed 35 HEALTHY frames while the measured current never left ±11 µA of its baseline. The 'collapse' was the

manual-trace → disc-fit method switch at the wrinkle transition — a change of measurement convention, not physics. All three observed collapses in the corpus carry a pair-mismatch note, and collapse corroborated exactly zero real breakdowns. Do not 'simplify' this back to an area- or absolute-μA-triggered rename. Collapse now confirms only alongside a current event at the same nominal-kV level; uncorroborated it becomes a note, 'collapse? area −N % (no current signature)'.

- Branding is symmetric and self-healing: rows at or after the first confirmed flag are branded (file renamed, frame_file updated, post-breakdown note); rows OUTSIDE that zone — including every row when the flags are empty, and rows never re-reviewed — are UN-branded. Files are renamed, never deleted, and a mismatch in either direction repairs itself on the next Save.
- Saved brands are not ground truth: the cross-run plot tool RECOMPUTES its marks from the saved current trace at plot time, a saved brand the recompute does not confirm gets a stale-brand warning and no mark, and no row is ever dropped for carrying a breakdown note.

The watchdog is a different instrument

The live watchdog runs during the run, samples the Trek current at ~2 Hz, and exists to STOP it. It is deliberately slow to trip: breakdown is declared only after the current has stayed at or above the trip level for confirm_s seconds of CONSECUTIVE samples. Any single reading below the threshold resets the streak; unreadable samples are ignored without resetting; the scope's 9.9E37 off-screen sentinel counts as an over-trip sample rather than an unreadable one, because a clipping current is a huge current, not a missing one. With a baseline learned at 0 kV before the ramp, the trip is on deviation, for the same −16 μA reason as above.

The post-hoc rule runs at Save, over the whole run at once, and exists to LABEL. It can see recovery, adjacency and the run's own median — none of which the watchdog has when it must decide. That is why a run can end without the watchdog firing and still carry a confirmed breakdown: 233451's −207 μA staircase was exactly that.

- Since 2026-08-05 the watchdog's samples go to telemetry.csv rather than being discarded, and every blank cell carries a status saying WHY it is blank (ok / offscreen / invalid / error / skipped) — an empty cell recording no reason is what made an earlier dropout investigation unresolvable. Telemetry is a record, not a safety device, and the code enforces that order: in a tick the watchdog decides FIRST and the row is written after.
- Stated rather than glossed: the sidecar was tested against a fake scope and is NOT bench-verified. Nothing in the analysis chain reads it, so no area, scale, verdict or breakdown flag can move because of it. Sampling is a 2 Hz MEAN, so a millisecond-scale arc still falls between samples — and no run in the 2026-07/08 batch has telemetry at all, so those runs cannot be dated electrically.

Labels, pairing and IoU

Hand traces are the ground truth this chain is validated against, and a trace is only useful as ground truth if it is PAIRED — stored alongside the machine candidate it should be compared with. A label with no machine candidate yields no IoU, ever.

That used to fail silently: opening an already-reviewed run without --auto means detection never ran in that session, so the candidate list is empty and the label went out with machine: null and no complaint. Four labels from the batch control round were lost that way. The fix CREATES the pairing rather than reporting its absence — when nothing has detected the current frame, the tracer detects that one frame before it opens — and a label can no longer be written with an unexplained missing pairing at all, because the record builder refuses unless the reason is named.

- A run-pass pairing comes from a full ▶ Detect Edges pass: ramp hysteresis applied, same-kV pairs reconciled. That is the convention the calibration curve is read in. A frame-on-demand pairing is one frame detected in isolation: no previous frame, so no hysteresis bonus, and pair reconciliation never runs.
- The on-demand caveats are quantified, not hand-waved. The missing 0.05 bonus is applied BEFORE candidates are sorted, so its absence can change WHICH candidate ranks first — measured 2026-08-06 on synthetic diff-tier scenes, the best area moved 3–9 % where the disc fit refuses, which is precisely the event and breakdown frames an operator hand-traces. So the conf can read up to 0.05 low AND a different candidate can rank first: honest as a pairing (stored conf and stored contour come from the same candidate), not a review result. Such candidates live in a separate store and never enter the review pass, and the trace report marks their points per conf bin and per method with a footnote naming the run and row.
- A REJECTED candidate still pairs: recovery — tracing when every automated candidate has been thrown out — is the tracer's first job, and the rejected candidate is exactly the machine answer the polygon is ground truth against. Two causes of an unpaired label survive on purpose: an unreadable baseline, and an honest no-change frame. Both get a dialog before the tracing effort, and the label carries the reason so the report names it. A trace whose pairing is impossible is still SAVED — recovery outranks calibration.
- What the calibration table means: P(IoU ≥ 0.8) per confidence bin plus per-method medians, over every label with a comparable machine candidate. It is the curve that decides what accept_conf may rise to, and its 0.8 target is validated against the measured human ceiling of 0.973.
- Vocabulary, and it has bitten. 'Row' means the data.csv row counted from 0, which is what the trace report means by it. Edge Review's status bar numbers FRAMES 1..N over the rows that HAVE a frame file, not over every CSV row, so one row without a frame shifts every later frame number by one — and an aborted run leaves exactly that. On 2026-08-07 an operator sent to 'row 28' navigated to frame 28 and landed the label on row 27, a valid but unintended frame. The report now prints both, 'row 28 (GUI frame 29)'.

Where to read next

- SLDEA_MEASUREMENT.md — the full error budget: §1.1 what uncertainty to quote for which quantity, §1.2 how much to trust each method, §2 every term with its measured size, §4 a provenance row for every number. Read it before quoting any figure in a methods section.
- SLDEA_HANDOFF.md — the dated, append-only decision log for the measurement chain. Every behaviour in this chapter has an observation → decision entry there with the evidence that settled it.
- CLAUDE.md — the timeless traps, including the two that cost the most time: the pre-2026-07-28 scale bug, and the calibration-method vocabulary.
- python sldea_trace.py <run-or-parent> — pools every edge_labels.json it finds and prints the conf-vs-IoU calibration table, plus a warning naming any label that can never yield an IoU. Any ranking change is re-scored against the stored polygons offline, at no operator cost.
- python sldea_diag.py <run> --out DIR — the per-run diagnostic, read-only. ALWAYS pass --out: run folders already hold outputs from earlier sessions and the default path overwrites them. And read sldea_diag_contact.png, not only the numbers — the statistics said nothing while every outline sat on the electrode strips.

Built from the live app (v1.2.0+8d86bcb) — every screenshot is a real capture, every callout is anchored to the actual control. Sources: `README.md`, `SLDEA_HANDOFF.md`, `SLDEA_MEASUREMENT.md` and the code itself.